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WORKLOAD SCHEDULING IN EDGE COMPUTING

Abstract

This disclosure provides methods, components, devices and systems for workload scheduling in edge computing. Some aspects more specifically relate to workload scheduling in distributed computing scenarios, which may include various devices including one or more workload scheduler devices, one or more host devices, and one or more client devices. A workload scheduler device may receive a workload request from a client device and may assign the workload to a host device from a set of host devices, with such a set of host devices including host devices capable of satisfying expectations of the workload. To assign the workload, the workload scheduler device may perform joint evaluations for the set of host devices, obtain scores for the set of host devices in accordance with the joint evaluations, rank the set of host devices in accordance with their respective scores, and assign the workload to the host device having the greatest ranking.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to wireless communication, and more specifically, to workload scheduling in edge computing.

DESCRIPTION OF THE RELATED TECHNOLOGY

[0002] Wireless communication systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (such as time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM). A wireless multiple-access communication system may include one or more base stations (BSs) or one or more network access nodes, each simultaneously supporting communication for multiple communication devices, which may be otherwise known as user equipment (UE).

SUMMARY

[0003] The systems, methods, and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for the desirable attributes disclosed herein.

[0004] One innovative aspect of the subject matter described in this disclosure can be implemented in a workload scheduler device. The workload scheduler device may include a processing system that includes processor circuitry and memory circuitry that stores code. The processing system may be configured to cause the workload scheduler device to receive, via a request associated with a workload, information indicative of one or more parameters associated with the workload and to transmit an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0005] Another innovative aspect of the subject matter described in this disclosure can be implemented in a method for connected edge workload scheduling by or at a workload scheduler device. The method may include receiving, via a request associated with a workload, information indicative of one or more parameters associated with the workload and transmitting an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0006] Another innovative aspect of the subject matter described in this disclosure can be implemented in a workload scheduler device. The workload scheduler device may include means for receiving, via a request associated with a workload, information indicative of one or more parameters associated with the workload and means for transmitting an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first

host device, and a first set of one or more reliability metrics associated with the first host device.

[0007] Another innovative aspect of the subject matter described in this disclosure can be implemented in a non-transitory computer-readable medium storing code for connected edge workload scheduling by or at a workload scheduler device. The code may include instructions executable by a processing system (such as one or more processors) to receive, via a request associated with a workload, information indicative of one or more parameters associated with the workload and transmit an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0008] Some implementations of the method, workload scheduler devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the first host device, information associated with the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0009] Some implementations of the method, workload scheduler devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for obtaining a set of scores for a set of host devices including the first host device in accordance with a respective joint evaluation associated with each host device of a set of host devices, where each score of the set of scores may be for a respective host device of the set of host devices, and where the assignment of the workload to at least the first host device may be in accordance with a first score for the first host device being relatively greater than one or more scores for a remainder of the set of host devices excluding the first host device.

[0010] Some implementations of the method, workload scheduler devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for obtaining a set of host devices from a set of multiple host devices in association with evaluating each host device of the set of multiple host devices in accordance with the one or more parameters and one or more variants of the workload, where the set of host devices includes the first host device.

[0011] In some implementations of the method, workload scheduler devices, and non-transitory computer-readable medium described herein, evaluating each host device of the set of multiple host devices may include operations, features, means, or instructions for evaluating whether the workload may be feasible for a respective host device of the set of multiple host devices in accordance with the one or more parameters, a respective set of one or more variants of the workload, and a capability of the respective host device.

[0012] Another innovative aspect of the subject matter described in this disclosure can be implemented in a first host device. The first host device may include a processing system that includes processor circuitry and memory circuitry that stores code. The processing system may be configured to cause the first host device to transmit, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device and receive, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters associated with the workload, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0013] Another innovative aspect of the subject matter described in this disclosure can be implemented in a method for connected edge workload scheduling by or at a first host device. The method may include transmitting, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device and receiving, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters associated with the workload, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0014] Another innovative aspect of the subject matter described in this disclosure can be implemented in a first host device. The first host device may include means for transmitting, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device and means for receiving, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters associated with the workload, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0015] Another innovative aspect of the subject matter described in this disclosure can be implemented in a non-transitory computer-readable medium storing code for connected edge workload scheduling by or at a first host device. The code may include instructions executable by a processing system (such as one or more processors) to transmit, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device and receive, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters associated with the workload, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0016] Some implementations of the method, first host devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the workload scheduler device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0017] Some implementations of the method, first host devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0018] Some implementations of the method, first host devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0019] Some implementations of the method, first host devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0020] Another innovative aspect of the subject matter described in this disclosure can be implemented in a client device. The client device may include a processing system that includes processor circuitry and memory circuitry that stores code. The processing system may be configured to cause the client device to transmit, via a request associated with a workload, information indicative of one or more parameters associated with the workload and receive information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0021] Another innovative aspect of the subject matter described in this disclosure can be implemented in a method for connected edge workload scheduling by a client device. The method may include transmitting, via a request associated with a workload, information indicative of one or more parameters associated with the workload and receiving information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0022] Another innovative aspect of the subject matter described in this disclosure can be implemented in client device. The client device may include means for transmitting, via a request associated with a workload, information indicative of one or more parameters associated with the workload and means for receiving information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0023] Another innovative aspect of the subject matter described in this disclosure can be implemented in a non-transitory computer-readable medium storing code for connected edge workload scheduling by or at a client device. The code may include instructions executable by a processing system (such as one or more processors) to transmit, via a request associated with a workload, information indicative of one or more parameters associated with the workload and receive information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0024] Some implementations of the method, client devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from a workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0025] Some implementations of the method, client devices, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from a workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0026] Details of one or more implementations of the subject matter described in this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 shows an example wireless communication system that supports workload scheduling in edge computing.

[0028] FIG. 2 shows an example connected edge cluster architecture that supports workload scheduling in edge computing.

[0029] FIG. 3 shows an example signaling diagram that supports workload scheduling in edge computing.

[0030] FIG. 4A shows an example workload scheduling diagram that supports workload scheduling in edge computing.

[0031] FIG. 4B shows an example nested workload scheduling algorithm that supports workload scheduling in edge computing.

[0032] FIG. 5 shows an example workload deployment algorithm that supports workload scheduling in edge computing.

[0033] FIG. 6 shows an example filtering procedure that supports workload scheduling in edge computing.

[0034] FIG. 7 shows an example scoring hierarchy that supports workload scheduling in edge computing.

[0035] FIG. 8 shows an example scoring adaptation algorithm that supports workload scheduling in edge computing.

[0036] FIG. 9 shows an example process flow that supports workload scheduling in edge computing.

[0037] FIGS. 10 and 11 show block diagrams of example devices that support workload scheduling in edge computing.

[0038] FIGS. 12-14 show flowcharts illustrating example methods that support workload scheduling in edge computing.

[0039] Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0040] The following description is directed to some implementations for the purposes of describing the innovative aspects of this disclosure. However, a person having ordinary skill in the art will readily recognize that the teachings herein can be applied in a multitude of different ways. The described implementations may be implemented in any device, system, or network that is capable of transmitting and receiving radio frequency (RF) signals according to any of the Institute of Electrical and Electronics Engineers (IEEE) 16.11 standards, or any of the IEEE 802.11 standards, the Bluetooth® standard, code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), Global System for Mobile communications (GSM), GSM/General Packet Radio Service (GPRS), Enhanced Data GSM Environment (EDGE), Terrestrial Trunked Radio (TETRA), Wideband-CDMA (W-CDMA),

Evolution Data Optimized (EV-DO), 1×EV-DO, EV-DO Rev A, EV-DO Rev B, High Speed Packet Access (HSPA), High Speed Downlink Packet Access (HSDPA), High Speed Uplink Packet Access (HSUPA), Evolved High Speed Packet Access (HSPA+), Long Term Evolution (LTE), AMPS, or other known signals that are used to communicate within a wireless, cellular or internet of things (IoT) network, such as a system utilizing third generation (3G), fourth generation (4G), fifth generation (5G), or sixth generation (6G), or further implementations thereof, technology.

[0041] Various aspects relate generally to assigning an edge compute workload to a host device. Some aspects more specifically relate to workload scheduling (such as workload assignments) in distributed computing scenarios, which may include various devices including one or more workload scheduler devices, one or more host devices, and one or more client devices. As described herein, “client” and “host” may be understood as logical roles of a device in a context of a workload assignment (such as a workload deployment). For example, a client device for a first workload may be a host device for a second workload. In other words, a device may serve, function, or operate as a host device or a client device on a per workload basis. In some implementations, a workload scheduler device may receive a workload request from a first device (which may be referred to herein as a client device for that workload request) and may assign the workload to a second device (which may be referred to herein as a host device for that workload request) for processing, computing, or execution. In some aspects, the workload scheduler device may assign the workload to the second device from a set of candidate devices (which may be referred to herein as a set of candidate host devices for that workload request). The workload scheduler device may obtain the set of candidate devices, which also may be referred to herein as a feasible host set, in accordance with determining, identifying, ascertaining, calculating, or otherwise receiving information indicating that each device of the set of candidate devices is capable of satisfying expectations associated with the requested workload. In other words, the workload scheduler device may obtain the feasible host set by filtering out potential host devices that are incapable of satisfying the expectations associated with the requested workload.

[0042] In accordance with obtaining the feasible host set, the workload scheduler device may perform one or more joint evaluations in association with assigning the requested workload. For example, the workload scheduler device may perform a respective joint evaluation for each device within the feasible host set. In other words, the workload scheduler device may perform a first joint evaluation for (such as associated with or corresponding to) a first device within the feasible host set, a second joint evaluation for (such as associated with or corresponding to) a second device within the feasible host set, and so on. Each joint evaluation may be a joint evaluation of one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics. For example, the first joint evaluation may be a joint evaluation of a first set of one or more computation metrics associated with the first device, a first set of one or more communication metrics associated with the first device, a first set of one or more power supply metrics associated with the first device, and a first set of one or more reliability metrics associated with the first device. Further, each joint evaluation may be associated with (such as informed or conditioned by) one or more parameters associated with the requested workload.

[0043] The workload scheduler device may obtain a respective score for each device within the feasible host set in accordance with performing the joint evaluations. For example, the workload scheduler device may obtain a first score for the first device in accordance with the first joint evaluation and may obtain a second score for the second device in accordance with the second joint evaluation. The workload scheduler device may rank the first device and the second device in accordance with the first score and the second score, respectively. In some implementations, the workload scheduler device may obtain the scores from the joint evaluations as weighted scores. In such implementations, the workload scheduler device may apply one or more weights to one or more of the computation metric(s), communication metric(s), power supply metric(s), and reliability metric(s) as part of each joint evaluation. For example, for each joint evaluation, the

workload scheduler device may apply a first weighting to the computation metric(s), a second weighting to the communication metric(s), a third weighting to the power supply metric(s), and a fourth weighting to the reliability metric(s). The first weighting, the second weighting, the third weighting, and the fourth weighting may be the same, partially the same, or different. For example, two or more of the weightings may be the same, two or more of the weightings may be different, or any combination thereof. In some aspects, the workload scheduler device may employ a learning capability or algorithm, such as via an artificial intelligence (AI) or machine learning (ML) model, to calculate, select, identify, or otherwise determine the weighting(s) for the joint evaluations. The workload scheduler device may provide such an AI or ML model with inputs of historical data associated with workload scheduling (such as via one or more workload logs) or user- or application-specific prioritizations (of, for example, one metric type over another), among other examples.

[0044] The workload scheduler device may assign the requested workload to a device from the feasible host set in accordance with the scores and the ranking of the devices within the feasible host set. For example, the workload scheduler device may assign the requested workload to the first device if the first score for the first device is relatively greater than a score for any other device within the feasible host set. In accordance with receiving an assignment of the requested workload, the first device (which may be a host device for the requested workload) may process the requested workload and return (such as transmit or otherwise provide) information associated with (such as a result of) the requested workload to the requesting device (which may be a client device for the requested workload).

[0045] Particular implementations of the subject matter described in this disclosure can be implemented to realize one or more of the following potential advantages. For example, by supporting some of the example implementations, a workload scheduler device may be able to more fully consider (such as account for, factor in, or evaluate) criteria that is especially relevant for distributed computing scenarios (such as distributed connected edge compute deployment scenarios). More specifically, by jointly evaluating one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics of each potential host device, the described techniques may be implemented to realize more dynamic, efficient, and suitable workload scheduling (such as assignment) as compared to other workload schedulers (as compared to workload schedulers unable to consider more than, for example, computation metrics). Such more dynamic, efficient, and suitable workload scheduling may relate to an assignment of a workload to a specific host device that is associated with a combination of computation capabilities, communication capabilities, power supply, and reliability that is relatively most compatible to a given set of parameters associated with the workload. For example, by considering reliability metrics (in addition to other metric types), the workload scheduler device may more frequently assign workloads to relatively more reliable devices, which may reduce the likelihood of an assigned workload being preempted at a host device (including in distributed computing scenarios associated with high dynamics and mobility). Further, by allocating workloads to suitable host devices considering the various metrics described herein, the described techniques may increase the likelihood of balancing resource usage across host devices and may support relatively more (such as a greater quantity of) workloads among a set of host devices (by way of more efficiently assigning workloads to different devices). Thus, the described techniques may increase the likelihood of an assigned host device being able to satisfy the parameters associated with the workload, which may support lower latency, higher data rates, service enhancements, and greater user experience, among other benefits.

[0046] FIG. 1 shows an example wireless communication system **100** that supports workload scheduling in edge computing. The wireless communication system **100** may include one or more network entities **105**, one or more UEs **115**, and a core network **130**. In some implementations, the wireless communication system **100** may be a Long Term Evolution (LTE) network, an LTE-

Advanced (LTE-A) network, an LTE-A Pro network, a New Radio (NR) network, or a network operating in accordance with other systems and radio technologies, including future systems and radio technologies not explicitly mentioned herein.

[0047] The network entities **105** may be dispersed throughout a geographic area to form the wireless communication system **100** and may include devices in different forms or having different capabilities. In various examples, a network entity **105** may be referred to as a network element, a mobility element, a RAN node, or network equipment, among other nomenclature. In some implementations, network entities **105** and UEs **115** may wirelessly communicate via one or more communication links **125** (such as a radio frequency (RF) access link). For example, a network entity **105** may support a coverage area **110** (such as a geographic coverage area) over which the UEs **115** and the network entity **105** may establish one or more communication links **125**. The coverage area **110** may be an example of a geographic area over which a network entity **105** and a UE **115** may support the communication of signals according to one or more radio access technologies (RATs).

[0048] The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communication system **100**, and each UE **115** may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. 1. The UEs **115** described herein may be capable of supporting communications with various types of devices, such as other UEs **115** or network entities **105**, as shown in FIG. 1.

[0049] As described herein, a node of the wireless communication system **100**, which may be referred to as a network node, or a wireless node, may be a network entity **105** (such as any network entity described herein), a UE **115** (such as any UE described herein), a network controller, an apparatus, a device, a computing system, one or more components, or another suitable processing entity configured to perform any of the techniques described herein. For example, a node may be a UE **115**. As another example, a node may be a network entity **105**. As another example, a first node may be configured to communicate with a second node or a third node. In one aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a UE **115**. In another aspect of this example, the first node may be a UE **115**, the second node may be a network entity **105**, and the third node may be a network entity **105**. In yet other aspects of this example, the first, second, and third nodes may be different relative to these examples. Similarly, reference to a UE **115**, network entity **105**, apparatus, device, computing system, or the like may include disclosure of the UE **115**, network entity **105**, apparatus, device, computing system, or the like being a node. For example, disclosure that a UE **115** is configured to receive information from a network entity **105** also discloses that a first node is configured to receive information from a second node.

[0050] In some implementations, network entities **105** may communicate with the core network **130**, or with one another, or both. For example, network entities **105** may communicate with the core network **130** via one or more backhaul communication links **120** (such as in accordance with an S1, N2, N3, or other interface protocol). In some implementations, network entities **105** may communicate with one another via a backhaul communication link **120** (such as in accordance with an X2, Xn, or other interface protocol) either directly (such as directly between network entities **105**) or indirectly (such as via a core network **130**). In some implementations, network entities **105** may communicate with one another via a midhaul communication link **162** (such as in accordance with a midhaul interface protocol) or a fronthaul communication link **168** (such as in accordance with a fronthaul interface protocol), or any combination thereof. The backhaul communication links **120**, midhaul communication links **162**, or fronthaul communication links **168** may be or include one or more wired links (such as an electrical link, an optical fiber link), one or more wireless links (such as a radio link, a wireless optical link), among other examples or various combinations thereof. A UE **115** may communicate with the core network **130** via a communication

link **155**.

[0051] One or more of the network entities **105** described herein may include or may be referred to as a base station (BS) **140** (such as a base transceiver station, a radio BS, an NR BS, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a gNB), a 5G NB, a next-generation eNB (ng-eNB), a Home NodeB, a Home eNodeB, or other suitable terminology). In some implementations, a network entity **105** (such as a BS **140**) may be implemented in an aggregated (such as monolithic, standalone) BS architecture, which may be configured to utilize a protocol stack that is physically or logically integrated within a single network entity **105** (such as a single RAN node, such as a BS **140**).

[0052] In some implementations, a network entity **105** may be implemented in a disaggregated architecture (such as a disaggregated BS architecture, a disaggregated RAN architecture), which may be configured to utilize a protocol stack that is physically or logically distributed among two or more network entities **105**, such as an integrated access backhaul (IAB) network, an open RAN (O-RAN) (such as a network configuration sponsored by the O-RAN Alliance), or a virtualized RAN (vRAN) (such as a cloud RAN (C-RAN)). For example, a network entity **105** may include one or more of a central unit (CU) **160**, a distributed unit (DU) **165**, a radio unit (RU) **170**, a RAN Intelligent Controller (RIC) **175** (such as a Near-Real Time RIC (Near-RT RIC), a Non-Real Time RIC (Non-RT RIC)), a Service Management and Orchestration (SMO) **180** system, or any combination thereof. An RU **170** also may be referred to as a radio head, a smart radio head, a remote radio head (RRH), a remote radio unit (RRU), or a transmission reception point (TRP). One or more components of the network entities **105** in a disaggregated RAN architecture may be co-located, or one or more components of the network entities **105** may be located in distributed locations (such as separate physical locations). In some implementations, one or more network entities **105** of a disaggregated RAN architecture may be implemented as virtual units (such as a virtual CU (VCU), a virtual DU (VDU), a virtual RU (VRU)).

[0053] The split of functionality between a CU **160**, a DU **165**, and an RU **170** is flexible and may support different functionalities depending on which functions (such as network layer functions, protocol layer functions, baseband functions, RF functions, and any combinations thereof) are performed at a CU **160**, a DU **165**, or an RU **170**. For example, a functional split of a protocol stack may be employed between a CU **160** and a DU **165** such that the CU **160** may support one or more layers of the protocol stack and the DU **165** may support one or more different layers of the protocol stack. In some implementations, the CU **160** may host upper protocol layer (such as layer 3 (L3), layer 2 (L2)) functionality and signaling (such as Radio Resource Control (RRC), service data adaptation protocol (SDAP), Packet Data Convergence Protocol (PDCP)). The CU **160** may be connected to one or more DUs **165** or RUs **170**, and the one or more DUs **165** or RUs **170** may host lower protocol layers, such as layer 1 (L1) (such as physical (PHY) layer) or L2 (such as radio link control (RLC) layer, medium access control (MAC) layer) functionality and signaling, and may each be at least partially controlled by the CU **160**. Additionally, or alternatively, a functional split of the protocol stack may be employed between a DU **165** and an RU **170** such that the DU **165** may support one or more layers of the protocol stack and the RU **170** may support one or more different layers of the protocol stack. The DU **165** may support one or multiple different cells (such as via one or more RUs **170**). In some implementations, a functional split between a CU **160** and a DU **165**, or between a DU **165** and an RU **170** may be within a protocol layer (such as some functions for a protocol layer may be performed by one of a CU **160**, a DU **165**, or an RU **170**, while other functions of the protocol layer are performed by a different one of the CU **160**, the DU **165**, or the RU **170**). A CU **160** may be functionally split further into CU control plane (CU-CP) and CU user plane (CU-UP) functions. A CU **160** may be connected to one or more DUs **165** via a midhaul communication link **162** (such as F1, F1 c, F1 u), and a DU **165** may be connected to one or more RUs **170** via a fronthaul communication link **168** (such as open fronthaul (FH) interface).

In some implementations, a midhaul communication link **162** or a fronthaul communication link **168** may be implemented in accordance with an interface (such as a channel) between layers of a protocol stack supported by respective network entities **105** that are in communication via such communication links.

[0054] In wireless communication systems (such as wireless communication system **100**), infrastructure and spectral resources for radio access may support wireless backhaul link capabilities to supplement wired backhaul connections, providing an IAB network architecture (such as to a core network **130**). In some implementations, in an IAB network, one or more network entities **105** (such as IAB nodes **104**) may be partially controlled by each other. One or more IAB nodes **104** may be referred to as a donor entity or an IAB donor. One or more DUs **165** or one or more RUs **170** may be partially controlled by one or more CUs **160** associated with a donor network entity **105** (such as a donor BS **140**). The one or more donor network entities **105** (such as IAB donors) may be in communication with one or more additional network entities **105** (such as IAB nodes **104**) via supported access and backhaul links (such as backhaul communication links **120**). IAB nodes **104** may include an IAB mobile termination (IAB-MT) controlled (such as scheduled) by DUs **165** of a coupled IAB donor. An IAB-MT may include an independent set of antennas for relay of communications with UEs **115**, or may share the same antennas (such as of an RU **170**) of an IAB node **104** used for access via the DU **165** of the IAB node **104** (such as referred to as virtual IAB-MT (vIAB-MT)). In some implementations, the IAB nodes **104** may include DUs **165** that support communication links with additional entities (such as IAB nodes **104**, UEs **115**) within the relay chain or configuration of the access network (such as downstream). In such implementations, one or more components of the disaggregated RAN architecture (such as one or more IAB nodes **104** or components of IAB nodes **104**) may be configured to operate according to the techniques described herein.

[0055] In the implementation of the techniques described herein applied in the context of a disaggregated RAN architecture, one or more components of the disaggregated RAN architecture may be configured to support workload scheduling in edge computing as described herein. For example, some operations described as being performed by a UE **115** or a network entity **105** (such as a BS **140**) may additionally, or alternatively, be performed by one or more components of the disaggregated RAN architecture (such as IAB nodes **104**, DUs **165**, CUs **160**, RUs **170**, RIC **175**, SMO **180**).

[0056] A UE **115** may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” also may be referred to as a unit, a station, a terminal, or a client, among other examples. A UE **115** also may include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some implementations, a UE **115** may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, or vehicles, meters, among other examples.

[0057] The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** that may sometimes act as relays as well as the network entities **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay BSs, among other examples, as shown in FIG. **1**.

[0058] The UEs **115** and the network entities **105** may wirelessly communicate with one another via one or more communication links **125** (such as an access link) using resources associated with one or more carriers. The term “carrier” may refer to a set of RF spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a portion of a RF spectrum band (such as a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a

given radio access technology (such as LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (such as synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communication system **100** may support communication with a UE **115** using carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers. Communication between a network entity **105** and other devices may refer to communication between the devices and any portion (such as entity, sub-entity) of a network entity **105**. For example, the terms “transmitting,” “receiving,” or “communicating,” when referring to a network entity **105**, may refer to any portion of a network entity **105** (such as a BS **140**, a CU **160**, a DU **165**, a RU **170**) of a RAN communicating with another device (such as directly or via one or more other network entities **105**).

[0059] Signal waveforms transmitted via a carrier may be made up of multiple subcarriers (such as using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may refer to resources of one symbol period (such as a duration of one modulation symbol) and one subcarrier, for which the symbol period and subcarrier spacing may be inversely related. The quantity of bits carried by each resource element may depend on the modulation scheme (such as the order of the modulation scheme, the coding rate of the modulation scheme, or both), such that a relatively higher quantity of resource elements (such as in a transmission duration) and a relatively higher order of a modulation scheme may correspond to a relatively higher rate of communication. A wireless communications resource may refer to a combination of an RF spectrum resource, a time resource, and a spatial resource (such as a spatial layer, a beam), and the use of multiple spatial resources may increase the data rate or data integrity for communications with a UE **115**.

[0060] The time intervals for the network entities **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, in some implementations, refer to a sampling period of $T_{\text{sub.s}} = 1/(\Delta f_{\text{sub.max}} \cdot N_{\text{sub.f}})$ seconds, for which $\Delta f_{\text{sub.max}}$ may represent a supported subcarrier spacing, and $N_{\text{sub.f}}$ may represent a supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (such as 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (such as ranging from 0 to 1023).

[0061] Each frame may include multiple consecutively numbered subframes or slots, and each subframe or slot may have the same duration. In some implementations, a frame may be divided (such as in the time domain) into subframes, and each subframe may be further divided into a quantity of slots. Alternatively, each frame may include a variable quantity of slots, and the quantity of slots may depend on subcarrier spacing. Each slot may include a quantity of symbol periods (such as depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communication systems **100**, a slot may further be divided into multiple mini-slots associated with one or more symbols. Excluding the cyclic prefix, each symbol period may be associated with one or more (such as $N_{\text{sub.f}}$) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.

[0062] A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (such as in the time domain) of the wireless communication system **100** and may be referred to as a transmission time interval (TTI). In some implementations, the TTI duration (such as a quantity of symbol periods in a TTI) may be variable. Additionally, or alternatively, the smallest scheduling unit of the wireless communication system **100** may be dynamically selected (such as in bursts of shortened TTIs (STTIs)).

[0063] Physical channels may be multiplexed for communication using a carrier according to

various techniques. A physical control channel and a physical data channel may be multiplexed for signaling via a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (such as a control resource set (CORESET)) for a physical control channel may be defined by a set of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (such as CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to an amount of control channel resources (such as control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE **115**.

[0064] In some implementations, a network entity **105** (such as a BS **140**, an RU **170**) may be movable and therefore provide communication coverage for a moving coverage area **110**. In some implementations, different coverage areas **110** associated with different technologies may overlap, but the different coverage areas **110** may be supported by the same network entity **105**. In some other implementations, the overlapping coverage areas **110** associated with different technologies may be supported by different network entities **105**. The wireless communication system **100** may include, for example, a heterogeneous network in which different types of the network entities **105** provide coverage for various coverage areas **110** using the same or different radio access technologies.

[0065] The wireless communication system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communication system **100** may be configured to support ultra-reliable low-latency communications (URLLC). The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions. Ultra-reliable communications may include private communication or group communication and may be supported by one or more services such as push-to-talk, video, or data. Support for ultra-reliable, low-latency functions may include prioritization of services, and such services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, and ultra-reliable low-latency may be used interchangeably herein.

[0066] In some implementations, a UE **115** may be configured to support communicating directly with other UEs **115** via a device-to-device (D2D) communication link **135** (such as in accordance with a peer-to-peer (P2P), D2D, or sidelink protocol). In some implementations, one or more UEs **115** of a group that are performing D2D communications may be within the coverage area **110** of a network entity **105** (such as a BS **140**, an RU **170**), which may support aspects of such D2D communications being configured by (such as scheduled by) the network entity **105**. In some implementations, one or more UEs **115** of such a group may be outside the coverage area **110** of a network entity **105** or may be otherwise unable to or not configured to receive transmissions from a network entity **105**. In some implementations, groups of the UEs **115** communicating via D2D communications may support a one-to-many (1:M) system in which each UE **115** transmits to each of the other UEs **115** in the group. In some implementations, a network entity **105** may facilitate the scheduling of resources for D2D communications. In some other implementations, D2D communications may be carried out between the UEs **115** without an involvement of a network entity **105**.

[0067] The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least

one control plane entity that manages access and mobility (such as a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (such as a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the network entities **105** (such as BSs **140**) associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to IP services **150** for one or more network operators. The IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

[0068] The wireless communication system **100** may operate using one or more frequency bands, which may be in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features, which may be referred to as clusters, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. Communication using UHF waves may be associated with smaller antennas and shorter ranges (such as less than 100 kilometers) compared to communications using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

[0069] The wireless communication system **100** may utilize both licensed and unlicensed RF spectrum bands. For example, the wireless communication system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology using an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. While operating using unlicensed RF spectrum bands, devices such as the network entities **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some implementations, operations using unlicensed bands may be in accordance with a carrier aggregation configuration in conjunction with component carriers operating using a licensed band (such as LAA). Operations using unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

[0070] A network entity **105** (such as a BS **140**, an RU **170**) or a UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a network entity **105** or a UE **115** may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more BS antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some implementations, antennas or antenna arrays associated with a network entity **105** may be located at diverse geographic locations. A network entity **105** may include an antenna array with a set of rows and columns of antenna ports that the network entity **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may include one or more antenna arrays that may support various MIMO or beamforming operations. Additionally, or alternatively, an antenna panel may support RF beamforming for a signal transmitted via an antenna port.

[0071] Beamforming, which also may be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (such as a network entity **105**, a UE **115**) to shape or steer an antenna beam (such as a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that some signals propagating along particular

orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (such as with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

[0072] The wireless communication system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or PDCP layer may be IP-based. An RLC layer may perform packet segmentation and reassembly to communicate via logical channels. A MAC layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer also may implement error detection techniques, error correction techniques, or both to support retransmissions to improve link efficiency. In the control plane, an RRC layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a network entity **105** or a core network **130** supporting radio bearers for user plane data. A PHY layer may map transport channels to physical channels.

[0073] The UEs **115** and the network entities **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly via a communication link (such as a communication link **125**, a D2D communication link **135**). HARQ may include a combination of error detection (such as using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (such as automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (such as low signal-to-noise conditions). In some implementations, a device may support same-slot HARQ feedback, for which the device may provide HARQ feedback in a specific slot for data received via a previous symbol in the slot. In some other implementations, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

[0074] Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices and may provide for automated communication between machines (such as via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a network entity **105** (such as a BS **140**) without human intervention. In some implementations, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay such information to a central server or application program that uses the information or presents the information to humans interacting with the application program. Some UEs **115** may be designed to collect information or enable automated behavior of machines or other devices. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

[0075] In some aspects, the wireless communication system **100** may support one or more signaling- or configuration-based mechanisms associated with workload scheduling in edge computing (such as in distributed computing scenarios). For example, the wireless communication system **100** may be an example of an edge computing system that includes a variety of devices, such as gNBs, APs, routers, mobile devices (smartphones), cars or other automobiles, laptops or other computing devices, gaming consoles, security cameras, wearable devices, extended reality (XR) or virtual reality (VR) glasses, augmented reality (AR) devices, smart thermostats, IoT devices, and other consumer devices connected to the network. These devices may serve as scheduler devices, host devices, client devices, or any two or more. Generally, a device may serve,

operate, or function as a host device or a client device on a per workload basis. For example, a device may serve as a client device for a first workload and may serve as a host device for a second workload (sequentially, such as at different times, or in parallel, such as simultaneously). Additionally, or alternatively, in comparison to host devices (such as edge nodes), client devices may have fewer computing resources and relatively limited processing capabilities (at least, for example, as relevant for a particular workload, such that a device may request assignment of workloads for which the device is relatively poorly suited or for which assignment would otherwise result in greater processing speed or device performance). Thus, in some scenarios, a client device may rely on a host device to process, analyze, and execute relatively larger computing workloads (such as processing tasks, task workloads, or data services) that would otherwise result in delays and power consumption at the client device. Some client devices, such as security cameras, may process workloads locally instead of offloading such workloads to other devices. In some implementations, a workload or processing task may be associated with an output, which can include data, commands (such as to increase the temperature of a thermostat), alerts (such as a police alert), notifications (such as to a homeowner), or exporting data to the cloud for analysis or logging (such as video footage), etc.

[0076] In some edge computing systems, the collective workload of the edge computing system may be distributed across various nodes (such as edge nodes or devices) in the edge computing system (such as UEs **115** and network entities **105**). Some edge computing systems may support compute resource management schemes for data centers, stationary compute nodes, and other devices. Such schemes may depend on communication links between devices having sufficient robustness and quality. However, these schemes may be unsuitable for mobile computing environments in which devices (such as cars, laptops, personal devices, and other compute nodes) move into the network, out of the network, or within the network at different times.

[0077] For example, in some mobile edge compute (MEC) deployments associated with cellular networks, a client device (such as a UE **115**) may move between different network entities **105**, resulting in changes to MEC latency criteria. Likewise, the topology of wireless communication links between host devices (such as edge nodes) and client devices may vary over time. Some edge computing deployments may be unable to account for device mobility when scheduling and distributing device workloads. For example, some transitory host devices (such as vehicles and automobiles, or mobile devices) may be unavailable during specific time periods and other host devices may be unable to adjust or redistribute task workloads according to the availability of transitory host devices. As described herein, a transitory host device may be a device or node with periodic, occasional, or intermittent availability, such that the transitory host device is temporarily available and otherwise unavailable for task processing or execution.

[0078] In some edge computing deployments, device workload distribution may be orchestrated (such as managed) by a cloud server or a data center. However, relying on cloud devices (such as a cloud edge **260** illustrated by and described with reference to FIG. **2**) for workload management and task execution may introduce delays, privacy concerns, and backhaul issues. For example, due to an inverse relationship between locality (such as distance) and latency (such as processing time) that exists in distributed computing environments, executing a processing workload at a nearby host device may be more effective (with respect to latency) than offloading the processing task to a remote cloud server. Thus, relying on the cloud to execute workloads may introduce unacceptable delays and security issues. Moreover, cloud devices may be several hops away from host devices or edge nodes within the edge computing system, and may experience connectivity issues. Further, cloud devices may consider a limited subset of a larger set of relevant criteria associated workload assignments in edge computing deployments, which may result in unsuitable or inefficient workload assignment. For example, while multiple different criteria may inform workload assignment in edge computing deployments, cloud devices may consider computation metrics and lack capability to consider other metrics pertinent to workload assignments in edge computing

deployments.

[0079] As an example, some schemes (such as Kubernetes schemes) associated with cloud workload orchestration may rely on cluster deployment over dedicated compute edge devices (which may not be available in some deployment scenarios) and may be unable to account for communication costs or a dynamic cluster topology (which may be especially impactful to overall performance in some deployment scenarios). Further, such schemes may lack non-containerized workload support and, in accordance with a reliance on dedicated compute edge devices, may not be suitable for heterogeneous (and non-Linux) device platforms. Additionally, such schemes may apply a heavy weight on edge devices with limited capabilities, which may be unsuitable or cause difficulties in scenarios in which edge devices are not exclusively available for edge computing tasks.

[0080] In some examples, host devices and client devices (such as one or more network entities **105** or one or more UEs **105**, or any combination thereof) of the wireless communication system **100** may support distributed computing scenarios and a corresponding scheduling (such as workload assignment) scheme. In other words, the wireless communication system **100** may be associated with a distributed computing ecosystem and may leverage a distributed computing workload scheduler. In accordance with supporting the distributed computing scenarios, host devices or client devices may bring (such as support) intelligence and high performance distributed computing to a connected edge. In some examples, the host devices or client devices may use a common (such as a same) platform to collect and process data and to learn an environment and its usage. Further, the host devices may manage resources and orchestrate tasks for efficiency and enhanced user experience. In some aspects, the host devices may leverage a hardware agnostic platform with open protocols, application programming interfaces (APIs), and software to enable the distributed computing ecosystem. Additionally, or alternatively, the host devices may use protocols, APIs, and algorithms configured for hardware of a specific vendor (such as a specific manufacturer).

[0081] Aspects of the present disclosure support distributed computing workload scheduling (which may be equivalently referred to as distributed computing workload orchestration) that addresses (such as considers, involves, factors in, or otherwise accounts for) edge characteristics. In other words, some example implementations relate to a distributed computing workload scheduler that more fully considers unique aspects of workload scheduling for distributed computing as compared to other schemes. For example, while workload scheduling in the cloud may focus exclusively on computation metrics, a workload scheduler device implementing the described techniques may assign a workload to a host device in accordance with a joint evaluation of one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics. In some implementations, the workload scheduler device may employ or otherwise use a workload deployment algorithm (such as a distributed computing workload deployment algorithm) to perform the joint evaluations of the computation metric(s), the communication metric(s), the power supply metric(s), and the reliability metric(s), as illustrated by and described in more detail with reference to FIG. 5.

[0082] Further, various aspects and techniques as described herein may be implemented, at least in part, using an AI program, such as a program that includes an ML or artificial neural network (ANN) model. An example ML model may include mathematical representations or define computing capabilities for making inferences from input data based on patterns or relationships identified in the input data. As used herein, the term “inferences” may include one or more of decisions, predictions, determinations, or values, which may represent outputs of the ML model. The computing capabilities may be defined in terms of one or more parameters of the ML model, such as weights and biases. Weights may indicate relationships between input data and outputs (such as between specific input data and specific outputs) of the ML model, and biases may be offsets that indicate a starting point for outputs of the ML model. An example ML model operating

on input data may start at an initial output based on (such as in accordance with or otherwise associated with) the biases and (subsequently) update its output based on a combination of the input data and the weights.

[0083] In some aspects, an ML model may be configured to provide computing capabilities for wireless communications. In some implementations, a workload scheduler device may use such an ML model to select one or more weightings (such as biases) for one or more joint evaluations (such as to flexibly and dynamically adjust relative priority levels between various metric types for which the workload scheduler device may account). Thus, during operation of a device, the workload scheduler device may use different weights or biases when scheduling different types of workloads. For example, the workload scheduler device may use different weights or biases when scheduling an ML-based workload as compared to other types of compute workloads.

[0084] ML models may be deployed in one or more devices (such as network entities **105** and UEs **115**) and may be configured to enhance various aspects of a wireless communication system. For example, an ML model may be trained to identify (or otherwise obtain) patterns or relationships in data corresponding to a network, a device, an air interface, or the like. An ML model may support operational decisions relating to one or more aspects associated with wireless communications devices, networks, or services. For example, an ML model may be utilized for supporting aspects such as workload assignments (including workload assignments in distributed computing scenarios), signal coding/decoding, network routing, energy conservation, transceiver circuitry controls, frequency synchronization, timing synchronization channel state estimation, channel equalization, channel state feedback, modulation, demodulation, device positioning, beamforming, load balancing, operations and management functions, or security, among other aspects.

[0085] ML models may be characterized in terms of types of learning that generate specific types of learned models that perform specific types of tasks. For example, different types of machine learning include supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. ML models may be used to perform different tasks such as classification or regression, where classification refers to determining (such as obtaining, predicting, selecting, identifying, or otherwise ascertaining) one or more discrete output values from a set of predefined output values, and regression refers to determining (such as obtaining, predicting, selecting, identifying, or otherwise ascertaining) continuous values which are not bounded by predefined output values. For example, a classification ML model configured according to aspects of this disclosure may produce an output that includes an assignment of a workload to a specific host device (from a set of available (such as predefined, preselected, or previously obtained) host devices). Some example ML models configured for performing such tasks include ANNs such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), transformers, diffusion models, regression analysis models (such as statistical models), large language models (LLMs), decision tree learning (such as predictive models), support vector networks (SVMs), and probabilistic graphical models (such as a Bayesian network). Further, it should be understood that, unless otherwise specifically stated, terms such “AI/ML model,” “ML model,” “trained ML model,” “ANN,” “model,” “algorithm,” or the like are intended to be interchangeable.

[0086] FIG. 2 shows an example connected edge cluster architecture **200** that supports workload scheduling in edge computing. The connected edge cluster architecture **200** illustrates communication between client devices **205**, host devices **210**, and a distributed computing controller **255** via network connectivity **250**. As described herein, a connected edge cluster (such as a cluster of devices associated with a distributed computing scenario) may be a group of locally connected edge devices (such as edge nodes) associated with a same administrator or user domain. A connected edge cluster may be equivalently referred to herein as a distributed computing local edge cluster, a distributed connected edge, or the like. In some implementations, one or more client devices **205**, one or more host devices **210**, or the distributed computing controller **255** may support distributed computing workload scheduling in accordance with one or more joint

evaluations of potential or candidate host devices **210**. In some aspects, a joint evaluation may be understood as being for (such as associated with or corresponding to) a given host device **210** and may be a joint evaluation of one or more computation metrics associated with that host device **210**, one or more communication metrics associated with that host device **210**, one or more power supply metrics associated with that host device **210**, and one or more reliability metrics associated with that host device **210**.

[0087] The client devices **205** may be examples of devices with relatively fewer processing capabilities or resources and the host devices **210** may be examples of devices with relatively more processing capabilities or resources. For example, the client devices **205** may be user devices, gaming consoles, gaming controllers or sensors, smart cameras, smart TVs, smart appliances (such as smart washers, dryers, refrigerators, freezers, microwaves, dishwashers, stoves, or ovens), XR or VR glasses or headsets, AR devices, smart watches, earbuds, smart thermostats, IoT devices, or any combination thereof. For further example, the host devices **210** may be laptops, mobile devices (smartphones), vehicles, servers, gNBs, APs, routers, or any combination thereof. Generally, devices such as gNBs, APs, UEs, mobile STAs, routers, mobile devices (smartphones), cars or other automobiles, laptops or other computing devices, gaming consoles, wearable devices, XR or VR glasses or headsets, AR devices, smart watches, earbuds, thermostats, sensors, IoT devices, and other consumer devices may serve as scheduler devices, host devices, client devices, or any two or more at different times. Further, although illustrated and described in examples as “host” devices and “client” devices, the various devices illustrated by and described with reference to FIG. **2** may serve, operate, or function as either a host device or a client device on a per workload basis.

[0088] The client devices **205** and the host devices **210** may be associated with (such as include, support, run, or otherwise perform operations via or in accordance with) one or more respective agents, components, functionalities, or applications. For example, a client device **205** may be associated with one or more third party applications **215** (such as third party distributed computing applications), one or more native applications **220** (such as native distributed computing applications), and a client agent **225** (such as a distributed computing client agent). A host device **210** may be associated with one or more third party applications **230** (such as third party distributed computing applications), one or more native applications **235** (such as native distributed computing applications), one or more workloads **240**, and a host agent **245** (such as a distributed computing host agent). In some aspects, the distributed computing controller **255**, which may include, function as, support, or otherwise be associated with a management and orchestration (MANO), may be an example of a workload scheduler device. In such aspects, the distributed computing controller **255** may perform one or more joint evaluations for (such as across, associated, or corresponding to) the set of host devices **210** and assign one or more workloads **240** to one or more host devices **210** in accordance with the one or more joint evaluations. The connected edge cluster architecture **200** may further include one or more legacy devices. In some aspects, such distributing computing components of the distributed computing controller **255**, the host devices **210**, the client devices **205**, and the legacy devices may differentiate distributed computing deployments from Kubernetes components of a master node (supporting or be associated with a scheduler and one or more controllers) and a worker node (supporting or associated with a kubelet, which may be a node-level agent that is in charge of executing pod targets, managing resources, and supporting cluster health).

[0089] The network connectivity **250** via which client devices **205**, host devices **210**, and the distributed computing controller **255** communicate (such as via over-the-air signaling) may include or be associated with any combination of Wi-Fi connectivity, cellular connectivity, Bluetooth connectivity, Zigbee connectivity, wired connectivity (such as via Ethernet), or IoT protocols such as Message Queuing Telemetry Transport (MQTT), Matter, ioFog, or other protocols. Client devices **205**, host devices **210**, and the distributed computing controller **255** may interface with network connectivity components via one or multiple interfaces. For example, a client device **205** and a host device **210** may support a distributed computing API **265** and a distributed computing

API **275**, respectively, and an original equipment manufacturer (OEM) API **270** and an OEM API **280**, respectively. The distributed computing controller **255** may support a distributed computing API **285** and, in some examples, may support a distributed computing API **290** to interface with a cloud edge **260**.

[0090] In some examples, the host devices **210** (which may be equivalently referred to herein as edge devices or edge nodes) may not be dedicated for running workloads **240** (such as computing workloads) assigned by the distributed computing controller **255** or another workload scheduler device. In other words, the host devices **210** may launch and run native applications **235** separate from, such as in addition to, any workloads **240** assigned to the host devices by the workload scheduler device (such as the distributed computing controller **255**). For example, the host devices **210** may be end user devices with on-device user applications outside the control of the workload scheduler device. In some aspects, a host device **210** and the distributed computing controller **255** may balance how many computing resources are available for (such as reserved for or used by) native (such as local or on-device) applications with how many computing resources are available for (such as reserved for or used by) assigned edge workloads **240**. Further, the host devices **210** (and the client devices **205**) may be on-premises devices with heterogeneous capabilities that may join or leave the cluster (such as the local edge cluster) or change connectivity within the cluster dynamically.

[0091] Thus, in accordance with some of the example implementations disclosed herein, the workload scheduler device may consider (such as account for, factor in, or evaluate) one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics to consider the relevant criteria more fully for distributed computing ecosystems in workload scheduling. For example, even though a first host device **210** may have relatively greater computation capabilities or resources as compared to a second host device **210**, the workload scheduler device may assign a workload to the second host device **210** if relatively greater communication metric(s), power supply metric(s), or reliability metric(s) of the second host device **210** sufficiently outweigh any shortcoming of the second host device **210** in terms of computation capabilities or resources. In such examples, the workload scheduler device may perform a first joint evaluation for the first host device **210** (of a first set of one or more computation metrics, communication metrics, power supply metrics, and reliability metrics associated with the first host device **210**) and a second joint evaluation for the second host device **210** (of a second set of one or more computation metrics, communication metrics, power supply metrics, and reliability metrics associated with the second host device **210**) and may assign the workload to the second host device **210** in accordance with the second joint evaluation providing (such as resulting in) a greater score than the first joint evaluation.

[0092] FIG. **3** shows an example signaling diagram **300** that supports workload scheduling in edge computing. The signaling diagram **300** may implement or be implemented to realize one or more aspects of the connected edge cluster architecture **200**. For example, the signaling diagram **300** illustrates communication between a client device **205-a**, a client device **205-b**, a host device **210-a**, a host device **210-b**, and a workload scheduler device **305**. The client device **205-a** and the client device **205-b** may be examples of client devices **205** as illustrated by and described with reference to FIG. **2**. The host device **210-a** and the host device **210-b** may be examples of host devices **210** as illustrated by and described with reference to FIG. **2**. Further, although illustrated and described in examples as “host” devices and “client” devices, the various devices illustrated by and described with reference to FIG. **3** may serve, operate, or function as either a host device or a client device on a per workload basis. The workload scheduler device **305** may be an example of a distributed computing controller **255** as illustrated by and described with reference to FIG. **2**. The client device **205-a**, the client device **205-b**, the host device **210-a**, the host device **210-b**, and the workload scheduler device **305** may communicate with each other via communication links **310**, which may be examples of over-the-air communication links (such as cellular, Bluetooth, or Wi-Fi links),

wired communication links (such as Ethernet links), or any combination thereof. The communication links **310** may be associated with the network connectivity **250** as illustrated by and described with reference to FIG. 2.

[0093] In some implementations, the client device **205-a** and the client device **205-b** may launch respective applications and, in accordance with launching the respective applications, may have respective workloads for processing. Additionally, or alternatively, the client device **205-a** and the client device **205-b** identify, select, determine, ascertain, or otherwise obtain respective workloads for processing in accordance with ongoing applications at the client device **205-a** and the client device **205-b**. In either or both of such scenarios, the client device **205-a** may have a first workload for processing and the client device **205-b** may have a second workload for processing.

[0094] In examples in which the client device **205-a** and the client device **205-b** are unequipped or lack the processing or computational capabilities or resources to process the first workload and the second workload, respectively (or otherwise determine or estimate that workload processing may be more efficiently or accurately performed by a different device), the client device **205-a** and the client device **205-b** may request assignment of the first and second workloads to one or more host devices **210**. In other words, the client device **205-a** and the client device **205-b** may request that host device(s) **210**, which may have relatively greater processing or computational capabilities or resources as compared to the client device **205-a** and the client device **205-b**, process the first and second workloads and return information associated with the workloads (such as results of the workloads) to the client device **205-a** and the client device **205-b** after processing. For example, the client device **205-a** may transmit a workload request **315-a** associated with the first workload and the client device **205-b** may transmit a workload request **315-b** associated with the second workload.

[0095] In some aspects, the client device **205-a**, via the workload request **315-a**, may provide information indicative of a first set of one or more parameters associated with the first workload. The client device **205-b**, via the workload request **315-b**, may provide information indicative of a second set of one or more parameters associated with the second workload. Such parameters associated with a requested workload may include one or more of a workload type, an indication of an affinity of the requested workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with a joint evaluation of a host device **210** for workload assignment, a requested service level agreement (SLA), or any combination thereof. The workload request **315-a** or the workload request **315-b** may be conveyed via a single message or packet or via multiple messages or packets.

[0096] A workload type may indicate that a requested workload is “session-oriented” or “one-shot.” A session-oriented workload type may indicate that a workload host is expected to keep (such as establish and maintain) a data session with a peer device (such as the requesting client device **205** or another device). For example, a session-oriented workload type may be associated with continuous, periodic, frequent, or ongoing processing tasks that are associated with data transmissions from the peer device. A one-shot workload type may indicate that one or more tasks associated with the workload run to completion on their own and that the input data is provided in the workload request (such as in the workload request **315-a** or workload request **315-b**). For example, a one-shot workload type may be completed without establishing a data session with a peer device.

[0097] An indication of an affinity of the requested workload may include information indicative of a compatibility or similarity between the requested workload and another workload or between the requested workload and a potential host device **210**. A requested completion time may be an indication of a time within which the requesting client device **205** expects a result associated with the requested workload. A requested power source or battery power metric may indicate a request for a host device **210** connected to an electrical grid (such as a “plugged-in” device) or may indicate a threshold battery power level (such that an assigned host device **210** may be expected to

have a battery power level that satisfies the threshold battery power level, if the host device **210** is not connected to the electrical grid).

[0098] A requested software or hardware capability may indicate a target operating system (OS), a target software or hardware architecture, a container flag, a quantity of central processing unit (CPU) cores, a quantity of graphics processing unit (GPU) cores, a storage availability, or a memory availability, among other examples of software or hardware that a client device **205** may request. One or more weightings may include weightings that the client device **205** requests the workload scheduler device **305** to use as part of a joint evaluation performed for workload assignment, such as in examples in which the workload scheduler device **305** uses the joint evaluation to obtain a weighted sum.

[0099] A requested SLA may indicate one or more criteria or metrics that the requesting client device **205** expects to be satisfied by the host device **210** assigned for workload processing, such as a quality of service (QoS). The QoS may be associated with a slice type, such as enhanced mobile broadband (eMBB), ultra-reliable low latency communication (URLLC), or massive machine type communication (mMTC). Additionally, or alternatively, a requested SLA may be workload type dependent. For example, for a session-oriented workload type, a requested SLA may indicate a first set of one or more metrics and, for a one-shot workload type, a requested SLA may indicate a second set of one or more metrics. Such a first set of one or more metrics may include one or more of a node ID for a session peer per direction (such as for peer-to-host communication and for host-to-peer communication), a target or expected throughput, or a target or expected latency. Such a second set of metrics may include an indication of a target or expected completion time.

[0100] The workload scheduler device **305**, which may be an edge node (such as an edge node with connectivity to a cloud edge, such as the cloud edge **260** as illustrated by and described with reference to FIG. 2), may receive the workload request **315-a** and the workload request **315-b** from the client device **205-a** and the client device **205-b**, respectively. In accordance with receiving the workload request **315-a** and the workload request **315-b**, the workload scheduler device **305** may assign the first and second workloads to one or more host devices **210**. For example, and as illustrated by FIG. 3, the workload scheduler device **305** may transmit an indication of an assignment **320-a** of the first workload to the host device **210-a** and may transmit an indication of an assignment **320-b** of the second workload to the host device **210-b**.

[0101] In some implementations, the workload scheduler device **305** may perform one or more joint evaluations **325** to determine or select (and transmit indications of) the assignment **320-a** to the host device **210-a** and the assignment **320-b** to the host device **210-b**. For example, the workload scheduler device **305** may perform a first set of joint evaluations **325** associated with (such as for) the first workload and may perform a second set of joint evaluations associated with (such as for) the second workload. The workload scheduler device **305** may perform the first set of joint evaluation **325** in accordance with performing a respective joint evaluation, for each host device **210** of a set of host devices **210** and in accordance with the first set of one or more parameters associated with the first workload, of one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics. Further, the workload scheduler device **305** may perform the second set of joint evaluation **325** in accordance with performing a respective joint evaluation, for each host device **210** of a set of host devices **210** and in accordance with the second set of one or more parameters associated with the second workload, of one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics.

[0102] For example, the first set of joint evaluations **325** may include a first joint evaluation **325** of the host device **210-a** for the first workload and a second joint evaluation **325** of the host device **210-b** for the first workload. In such examples, the first joint evaluation **325** of the host device **210-a** may consider (such as account for or factor in) a first set of one or more computation metrics associated with the host device **210-a**, a first set of one or more communication metrics associated

with the host device **210-a**, a first set of one or more power supply metrics associated with the host device **210-a**, and a first set of one or more reliability metrics associated with the host device **210-a**. The second joint evaluation **325** of the host device **210-b** may consider (such as account for or factor in) a second set of one or more computation metrics associated with the host device **210-b**, a second set of one or more communication metrics associated with the host device **210-b**, a second set of one or more power supply metrics associated with the host device **210-b**, and a second set of one or more reliability metrics associated with the host device **210-b**. The workload scheduler device **305** may obtain a first score from the first joint evaluation **325** and a second score from the second joint evaluation **325** and may assign the first workload to the host device **210-a** or the host device **210-b** in accordance with which of the first score or the second score is greater. In some aspects, the workload scheduler device **305** may assign the first workload to the host device **210-a** (via the assignment **320-a**) in accordance with the first score being greater than the second score. [0103] For further example, the second set of joint evaluations **325** may include a third joint evaluation **325** of the host device **210-a** for the second workload and a fourth joint evaluation **325** of the host device **210-b** for the second workload. In such examples, the third joint evaluation **325** of the host device **210-a** may consider (such as account for or factor in) a first set of one or more computation metrics associated with the host device **210-a**, a first set of one or more communication metrics associated with the host device **210-a**, a first set of one or more power supply metrics associated with the host device **210-a**, and a first set of one or more reliability metrics associated with the host device **210-a**. The fourth joint evaluation **325** of the host device **210-b** may consider (such as account for or factor in) a second set of one or more computation metrics associated with the host device **210-b**, a second set of one or more communication metrics associated with the host device **210-b**, a second set of one or more power supply metrics associated with the host device **210-b**, and a second set of one or more reliability metrics associated with the host device **210-b**. The workload scheduler device **305** may obtain a third score from the third joint evaluation **325** and a fourth score from the fourth joint evaluation **325** and may assign the second workload to the host device **210-a** or the host device **210-b** in accordance with which of the third score or the fourth score is greater. In some aspects, the workload scheduler device **305** may assign the second workload to the host device **210-b** (via the assignment **320-b**) in accordance with the fourth score being greater than the third score.

[0104] In some implementations, the workload scheduler device **305** may continuously track (such as receive indications of or otherwise monitor) resource availability or usage on host devices **210** and may incorporate resource availability or usage into the joint evaluations **325**. For example, the workload scheduler device **305** may set (such as select, determine, identify, calculate, or otherwise ascertain) a maximum (such as upper limit) cap on distributed computing usage **335** for each host device **210** to avoid adversely impacting native application performance. In other words, because the host devices **210** may be absent of (such as lack) dedicated resources for distributed computing applications, the workload scheduler device **305** may assign workloads to host devices **210** in view of a possibility that a host device **210** launches a native application while processing an assigned distributed computing workload. In some deployment scenarios or for some types of host devices **210**, non-distributed computing usage **340** may be highly dynamic depending on user interaction (such as a launching of a native application, such as a launching of a video game on a laptop).

[0105] In some aspects, the workload scheduler device **305** may set the limit (such as a cap) on distributed computing usage **335** such that distributed computing usage **335** is less than or equal to a threshold percentage (or ratio) of resources (such as integrated hardware resources, such as computational, memory, or storage resources) of the host device **210**. Such a threshold percentage may be set to x %, where x may be any quantity including 10, 15, 20, 25, 30, and so on. Additionally, or alternatively, the workload scheduler device **305** may set a limit on distributed computing usage **335** such that distributed computing usage **335** is associated with less than or equal to a threshold quantity of resources, where such a threshold quantity of resources may refer

to a quantity of processing blocks, a quantity of CPU cores, a quantity of GPU cores, an amount of memory, or the like. Further, in view of the highly dynamic nature of the non-distributed computing usage **340**, the workload scheduler device **305** may set a limit on distributed computing usage **335** to provide for an amount (such as a threshold percentage or a threshold quantity) of idle resources **330**. Thus, the workload scheduler device **305** may set a limit on distributed computing usage **335** such that distributed computing usage **335** is less than or equal to a threshold percentage or quantity, such that non-distributed computing usage **340** is at least a threshold percentage or quantity, such that idle resources are at least a threshold percentage or quantity, or any combination thereof. If, for example, computation, memory, or storage usage for distributed computing workloads exceeds a fixed percentage or quantity at a host device **210**, or if a power of the host device **210** falls below a fixed (or self-defined) percentage or ratio, a status of the host device **210** may be set to busy and the workload scheduler device **305** may skip the host device **210** for workload assignments. In some implementations, the workload scheduler device **305** may skip the host device **210** during host a filtering procedure, as illustrated by and described in more detail with reference to FIGS. **5** and **6**.

[0106] The workload scheduler device **305** may set a respective maximum cap on distributed computing usage **335** on each host device **210** in accordance with usage history of that host device **210**. In some implementations, the workload scheduler device **305** may set a same maximum cap on distributed computing usage **335** for each host device **210** of a set of host devices **210**. In some other implementations, the workload scheduler device **305** may set different (or at least separately defined or indicated) maximum caps on distributed computing usage **335** for each host device **210** of a set of host devices **210**. The workload scheduler device **305** may account for such resource availability or usage (in view of the caps on distributed computing usage **335**) as a computation or reliability metric or may account for such resource availability or usage as a separate metric (such as a standalone metric separate from, and in addition to, computation and reliability metrics).

[0107] Additionally, or alternatively, the workload scheduler device **305** may monitor for link utilization or packet loss reports. For example, APs or other network entities **105** may transmit (such as report), to the workload scheduler device **305**, information indicative of a link utilization or information indicative of a packet loss (such as retransmission), or both. In such examples, the workload scheduler device **305** may use such reported information to determine, identify, or ascertain one or more congested areas of a network (such as some hotspots of signaling congestion). The workload scheduler device **305** may determine, identify, or ascertain one or more congested areas in accordance with one or more metrics associated with a quantity of channel accesses, a percentage of available spectrum used, a quantity of collisions, a quantity of retransmissions, a quantity of scheduling requests, a quantity of connection (such as random access) attempts, or any combination thereof.

[0108] Additionally, or alternatively, the workload scheduler device **305** may monitor whether the parameters, targets, requests, or expectations of assigned workloads are satisfied. In other words, the workload scheduler device **305** may monitor workload execution at one or more host devices **210**. For example, the workload scheduler device **305** may receive, from client devices **205** or host devices **210**, information indicative of whether one or more parameters associated with assigned workloads are consistently satisfied (such as satisfied at least a threshold percentage of the time, or such that a threshold quantity or percentage of the parameters are satisfied). If the workload scheduler device **305** receives information indicating that one or more parameters associated with assigned workloads are not consistently satisfied, the workload scheduler device **305** may reschedule one or more workloads, reassign one or more workloads, or reconfigure a topology of the network.

[0109] Further, if the workload scheduler device **305** (re)deploys a sequence of workloads or receives information indicating that the sequence of workloads are being (re)deployed (such as at the workload scheduler device **305**, at a given host device **210**, or across a set of multiple host

devices **210**), the workload scheduler device **305** may calculate (or receive information indicative of) a weighted sum of the queuing times associated with the sequence of workloads. Such a weighted sum may be weighted in accordance with (such as based on) using workload priorities as weightings. For example, a first workload associated with a first queuing time and a first priority value may be given a weighted queuing time associated with (such as equal to) a first product of the first queuing time and the first priority value. For further example, a second workload associated with a second queuing time and a second priority value may be given a weighted queuing time associated with (such as equal to) a second product of the second queuing time and the second priority value. In such examples, a weighted sum of the queuing times may include a summation of the first product and the second product. A relatively large (weighted sum) queuing time (such as larger than or equal to a threshold time) may indicate a cluster status is busy. Likewise, a relatively small (weighted sum) queuing time (such as smaller than or equal to a threshold time) may indicate that a cluster status is idle or available.

[0110] In some implementations, the workload scheduler device **305** may monitor a distribution of workloads across a set of host devices **210** and may assign workloads to balance the distribution of the workloads across the set of host devices **210**. For example, the workload scheduler device **305** may assign workloads to the host device **210-a** and the host device **210-b** such that a first set of workloads assigned to the host device **210-a** is approximately commensurate to a second set of workloads assigned to the host device **210-b**. The first set of workloads may be approximately commensurate to the second set of workloads in accordance with the first set of workloads and the second set of workloads being associated with approximately equal quantities or percentages (measured on a per-device basis) of computation resources, approximately equal priorities, or any combination thereof.

[0111] Additionally, or alternatively, the workload scheduler device **305** may monitor a reliability of one or more host devices **210** in accordance with monitoring a rescheduling or a preemption rate at the one or more host devices **210**. For example, the workload scheduler device **305** may use a percentage of rescheduling or preemption to characterize a reliability of a cluster. A host device **210** may reschedule or preempt an assigned distributed computing workload in accordance with a network change, an exit of the host device **210** from the cluster, a local application launch, or a higher-priority edge compute workload launch.

[0112] The workload scheduler device **305** may perform one or more workload updates in accordance with distributed computing cluster monitoring (such as status monitoring of distributed computing clusters, such as in accordance with monitoring percentage usage of resources at host devices **210**, link utilization or packet loss reports, workload execution, workload distribution, or rescheduling/preemption). In some aspects, in accordance with host mobility, network condition(s), local applications, or higher-priority edge compute workloads at a host device **210**, the workload scheduler device **305** may redeploy an assigned workload or adjust, reconfigure, or otherwise update a network or cluster topology. Additionally, or alternatively, if the workload scheduler device **305** receives information indicating that one or more workloads are past a completion deadline, the workload scheduler device **305** may send the workload (or an indication of the workload, potentially along with an indication of a rejection or decline of the workload) back to the client device **205**. In some examples, the workload scheduler device **305** or a host device **210** may sort (such as order or organize) workloads in accordance with a respective priority or a respective completion deadline associated with each of the workloads. In some examples, the workload scheduler device **305** may (re)assign workloads or (re)configure a workload deployment to balance a workload distribution across a set of host devices **210** (or in accordance with receiving information indicating that a workload distribution is imbalanced). Further, in some aspects, the workload scheduler device **305** may limit a quantity of workloads in a queue. For example, the workload scheduler device **305** may set (such as configure) a maximum or upper limit quantity of N workloads allowed in a queue of the workload scheduler device **305**.

[0113] Further, in terms of distributed computing usage **335** at a host device **210** (such as the host device **210-a**), the workload scheduler device **305** or the host device **210** may perform one or more operations in accordance with whether one or more threshold percentages are satisfied. For example, if the distributed computing usage **335** at the host device **210-a** satisfies (such as exceeds) a first threshold percentage $x_{sub.1}$, the workload scheduler device **305** may skip the host device **210-a** in a filtering step. Additionally, or alternatively, if the distributed computing usage **335** at the host device **210-a** satisfies (such as exceeds) a second threshold percentage $x_{sub.2}$, the host device **210-a** may preempt one or more workloads assigned to the host device **210-a** (in favor of allocating sufficient resources for one or more native applications of the host device **210-a**). Additionally, or alternatively, if the distributed computing usage **335** at the host device **210-a** satisfies (such as exceeds) a third threshold percentage $x_{sub.3}$, the host device **210-a** may reject new incoming workloads. For example, the host device **210-a** may transmit an indication of a rejection to the workload scheduler device **305** in association with receiving an assignment of a workload from the workload scheduler device **305** when the distributed computing usage **335** at the host device **210-a** satisfies (such as exceeds) the third threshold percentage $x_{sub.3}$.

[0114] In accordance with monitoring link utilization, packet loss reports, or rescheduling/preemption occurrences, the workload scheduler device **305** may selectively prohibit or reject workload requests. For example, if the network becomes congested (such as in accordance with or otherwise indicated by a relatively high packet loss, retransmission, or link utilization), the workload scheduler device **305** may stop (all or relatively lower priority) incoming session-oriented workloads (such as session-oriented jobs). In such examples, the workload scheduler device **305** may transmit an indication of a rejection of (such as a prohibition against) workloads associated with a session-oriented workload type. Additionally, or alternatively, if a rescheduling/preemption rate is relatively high (such as greater than or equal to a threshold rate), the workload scheduler device **305** may reject new incoming workload requests until the distributed computing cluster is stable again. In such examples, the workload scheduler device **305** may transmit an indication of a rejection of (such as a prohibition against) workloads associated any workload type.

[0115] In some implementations, the workload scheduler device **305** may support a compute aware network resource allocation. In such implementations, the workload scheduler device **305**, or one or more host devices **210** or other edge nodes, may offload links on highly utilized channels to other channels. For example, the workload scheduler device **305**, or another edge node, may change a connection between two network devices (such as APs or any other network entities **105**) from a first channel (such as a 2.4 GHz channel) to a second channel (such as a 5 GHz channel). Additionally, or alternatively, the workload scheduler device **305** may configure a trajectory for workloads in accordance with a network resource allocation. For example, the workload scheduler device **305** may change a route of one or more session-oriented workloads to one or more links with relatively less traffic. Additionally, or alternatively, the workload scheduler device **305** may limit the quantity of incoming session-oriented workloads (such as to an upper limit threshold quantity) in distributed computing clusters to avoid causing network congestion.

[0116] Additionally, or alternatively, the workload scheduler device **305** may steer a host device **210** to a specific network device (such as a specific AP, gNB, router, or other network entity **105**) if there are distributed computing applications running on the specific network device. In other words, the workload scheduler device **305** may connect the host device **210** with a network device having a better connection for the host device **210** (such as a distributed computing host), such as a connection that is relatively better for a particular workload. In such examples, the workload scheduler device **305** may transmit, to the host device **210**, a first network device, or a second network device, information indicative of a handover of the host device **210** to the first network device from the second network device. In some aspects, the host device **210** may have a wired connection (such as an Ethernet connection) with one or both of the first network device or the second network device. In such aspects, the workload scheduler device **305** may steer the host

device **210** to a wired connection from a wireless connection, to a wireless connection from a wired connection, to a first wired connection from a second wired connection, or to a first wireless connection from a second wireless connection. Further, in some implementations, the workload scheduler device **305** may disconnect other devices connected to the first network device. Additionally, or alternatively, the first network device may change a (dedicated) channel for the host device **210** (such as the compute node). In some examples, the workload scheduler device **305** may configure network devices (functioning as or supporting host devices **210**) to serve at most one or a fixed quantity of session-oriented workloads per channel. In such examples, the workload scheduler device **305** may transmit information indicative of such a fixed quantity to the network devices. Additionally, or alternatively, such a fixed quantity may be provided by a network specification.

[0117] FIG. **4A** shows an example workload scheduling diagram **400** that supports workload scheduling in edge computing. The workload scheduling diagram **400** may implement or be implemented to realize one or more aspects of the connected edge cluster architecture **200** or the signaling diagram **300**. For example, the workload scheduling diagram **400** illustrates an interfacing between a client application **405**, a workload scheduler device **305**, and a host device **210**. The host device **210** may be an example of a host device **210** as illustrated by and described with reference to FIGS. **2** and **3** and the workload scheduler device **305** may be an example of a workload scheduler device **305** as illustrated by and described with reference to FIG. **3**. The client application **405** may run at a client device **205**, such as a client device **205** as illustrated by and described with reference to FIGS. **2** and **3**.

[0118] The workload scheduler device **305** may interface with the client application **405** via a workload scheduling API **410** and may interface with the host device **210** via a workload deployment API **415**. Further, although illustrated as having one workload scheduling API **410** and one workload deployment API **415**, the workload scheduler device **305** may have any quantity of workload scheduling APIs **410** (such as one for each client application **405** of a set of client applications **405**) and any quantity of workload deployment APIs **415** (such as one for each host device **210** of a set of host devices **210**). The workload scheduling API **410** may be an external workload orchestration API for distributed computing client applications. The workload deployment API **415** may be an example of an internal workload deployment API to the host devices **210** (via, for example, Device Service).

[0119] The workload scheduler device **305**, in accordance with aspects of the workload scheduling diagram **400**, may manage containerized and native application workloads within a distributed computing cluster, including workload scheduling (such as finding the right host device **210** for each application workload) and workload lifespan management. In some implementations, the workload scheduler device **305** (functioning as a distributed computing workload orchestrator) may orchestrate workloads between the edge and the cloud. For example, if (sufficient) resources are not available on the edge, the workload scheduler device **305** may offload some workloads to the cloud (such as the cloud edge **260** as illustrated by and described with reference to FIG. **2**). Thus, and in accordance with some example implementations disclosed herein, the workload scheduler device **305** may leverage cloud capabilities while addressing edge-specific characteristics.

[0120] FIG. **4B** shows an example nested workload scheduling algorithm **401** that supports workload scheduling in edge computing. The nested workload scheduling algorithm **401** may implement or be implemented to realize one or more aspects of the connected edge cluster architecture **200**, the signaling diagram **300**, or the workload scheduling diagram **400**. For example, a workload scheduler device **305**, such as a workload scheduler device **305** as illustrated by and described with reference to FIGS. **3** and **4A**, may perform one or more operations associated with the nested workload scheduling algorithm **401** as part of distributed computing workload scheduling.

[0121] In accordance with the nested workload scheduling algorithm **401**, the workload scheduler

device **305** or another edge node may perform one or more operations association with a resource configuration **420** (such as a resource optimization), a workload configuration **425** (such as a workload optimization), and a workload deployment **430**. The resource configuration **420** may include a (re) configuring, by the workload scheduler device **305** or another edge node, of a cluster resource. For example, the workload scheduler device **305** or another edge node may update a topology associated with a distributed computing cluster. The workload configuration **425** may include a (re) deploying, by the workload scheduler device **305** or another edge node, of one or more active workloads. For example, the workload scheduler device **305** or another edge node may redeploy one or more active workloads in accordance with a preemption or local/global bin packing.

[0122] The workload deployment **430** may include a selecting, by the workload scheduler device **305**, of a host device **210** (such as a host device **210** as illustrated by and described with reference to FIGS. **2**, **3**, and **4A**) to which to deploy a new workload in accordance with a client request. For example, the workload scheduler device **305** may transmit an indication of assignment of a workload to at least a host device **210** as part of the workload deployment **430**. In accordance with some example implementations disclosed herein, the workload scheduler device **305** may assign a workload to a host device **210** in accordance with a respective joint evaluation, for each host device **210** of a set of host devices **210** and in accordance with one or more parameters associated with the workload or a corresponding client request, of one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics.

[0123] In some examples, algorithm complexity, cluster performance, or execution cost may scale in correlation with a progression from the workload deployment **430** to the workload configuration **425** and to the resource configuration **420**. Further, the nested workload scheduling algorithm **401** may be associated with a set of one or more inputs and a set of one or more outputs. The set of one or more inputs may include, for example, a cluster resource specification (such as a resource configuration or allocation), an existing workload-to-host device mapping, or a workload specification (such as one or more parameters associated with a workload). The set of one or more outputs may include, for example, an updated workload-to-host device mapping or an updated cluster resource specification.

[0124] FIG. **5** shows an example workload deployment algorithm **500** that supports workload scheduling in edge computing. The workload deployment algorithm **500**, which may be an example of a distributed computing workload deployment algorithm, may implement or be implemented to realize aspects of the wireless communication system **100**, the connected edge cluster architecture **200**, the signaling diagram **300**, the workload scheduling diagram **400**, or the nested workload scheduling algorithm **401**. For example, a workload scheduler device **305** may perform the operations associated with the workload deployment algorithm **500** and such a workload scheduler device **305** may be an example of a distributed computing controller **255** or a workload scheduler device **305** as illustrated by or described with reference to FIG. **2**, **3**, **4A**, or **4B**.

[0125] The workload deployment algorithm **500** may be associated with a three-step process involving a first step of looping through one or multiple workload variants while finding feasible host devices **210**, a second step of finding a set of host devices **210** capable of running the workload, and a third step of ranking the feasible (such as eligible) host devices **210** to select a best host device **210** for the workload, with such a “best” host device **210** being understood as a host device **210** having a greatest score, a lowest score, or a score closest to a target score in accordance with the ranking. In some aspects, the first step may be understood as a multiple variant step, the second step may be understood as a filtering step, and the third step may be understood as a scoring step.

[0126] The workload scheduler device **305** may initiate, start, begin, or trigger the workload deployment algorithm **500** in association with receiving a workload request **505**. In some examples, the workload scheduler device **305** may receive the workload request **505**, which may be a request

for or otherwise associated with a workload, from a client device **205**. The workload request **505** may be an example of the workload request **315-a** or the workload request **315-b** as illustrated by and described with reference to FIG. 3. In some aspects, the workload scheduler device **305** may receive, via the workload request **505** or via separate signaling or messaging, one or more parameters associated with the workload. Such parameters associated with a requested workload may include one or more of a workload type, an indication of an affinity of the requested workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with a joint evaluation of a host device **210** for workload assignment, a requested SLA, or any combination thereof.

[0127] The workload scheduler device **305**, in accordance with receiving the workload request **505** and the one or more parameters associated with the workload, may obtain (such as identify, configure, calculate, or generate) a set of one or more workload variants associated with the workload and may evaluate a host device **210** to identify, calculate, determine, or ascertain whether the host device **210** is capable of processing any one or more of the various workload variants. Such various workload variants may include, for purpose of example in the illustration of FIG. 5, one or more of a workload variant **510-a**, a workload variant **510-b**, or a workload variant **510-c**. Each workload may be associated with a specific permutation of a set of workload factors. Such workload factors may include one or more of an image URI, an architecture (such as x86, arm64, RISC-V, or the like), an operating system (such as Linux, Windows, Android, or the like), a container flag (such as “true” or “false”), or resource expectations or requirements (such as for storage, memory, CPU capability (in units of, for example, millicpu), or GPU capability). In some aspects, the workload scheduler device **305**, the client device **205**, or the host device **210** may support different variants of workloads to fit into (such as to be compatible with) different device platforms and capabilities.

[0128] The workload scheduler device **305** may perform a filtering **515** to identify, determine, or ascertain whether a host device **210** is capable of processing at least one of the various workload variants. For example, if a host device **210** is capable of processing at least one of the various workload variants, the workload scheduler device **305** may add the host device **210** to a feasible host set (such as a set of host devices **210**). If the host device **210** is not capable of processing at least one of the various workload variants, the workload scheduler device **305** may exclude the host device **210** from the feasible host set. The workload scheduler device **305** may perform the filtering **515** across all potential host devices **210** (in view of at least one of the various workload variants) to obtain the feasible host set, which may include one or multiple host devices **210**. In some aspects, the workload scheduler device **305** may start by performing the filtering **515** using a first (such as highest priority) workload variant and, if no host devices **210** are capable of processing the first workload variant, may subsequently perform the filtering **515** using a second (such as next highest priority) workload variant, and so on until the workload scheduler device **305** finds a workload variant supported by at least one host device **210**. Additional details relating to the filtering **515** are illustrated by and described with reference to FIG. 6.

[0129] For example, in accordance with the filtering **515** to obtain the feasible host set, the workload scheduler device **305** may perform a decision **520** to identify, determine, or ascertain whether the feasible host set is empty. An empty feasible host set may indicate that the feasible host set includes zero host devices **210**. If the feasible host set is empty, meaning that all potential host devices **210** were filtered out by the filtering **515**, the workload scheduler device **305** may perform a decision **525** to identify, determine, or ascertain whether a most recently evaluated workload variant is a last workload variant (of a set of generated workload variants). If not, the workload scheduler device **305** may evaluate a next workload variant to identify, determine, or ascertain whether one or more host devices **210** are capable of processing the next workload variant. If yes, the workload scheduler device **305** may perform a workload configuration **530**. The workload configuration **530**, which may be alternatively referred to as a workload optimization, may be an

example of the workload configuration **425** as illustrated by and described with reference to FIG. **4B**. The workload configuration **530** (and, likewise, the workload configuration **425**) may be understood as an outer loop or a second phase of a nested workload scheduling algorithm, such as the nested workload scheduling algorithm **401** as illustrated by and described with reference to FIG. **4B**.

[0130] Alternatively, if the workload scheduler device **305** identify, determines, or ascertains that the feasible host set is not empty at the decision **520**, the workload scheduler device **305** may perform a scoring **535** for each host device **210** within the feasible host set. In accordance with the scoring **535**, the workload scheduler device **305** may obtain (such as calculate, determine, select, or ascertain) a score or rank for each host device **210** within the feasible host set. For example, if the feasible host set includes a first host device **210** and a second host device **210**, the workload scheduler device **305** may obtain, via the scoring **535**, a first score or rank for the first host device **210** and a second score or rank for the second host device **210**. Additional details relating to the scoring **535** are illustrated by and described with reference to FIG. **7**.

[0131] In accordance with the scoring **535**, the workload scheduler device **305** may perform a workload assignment **540** to deploy the workload on a selected host device **210**. For example, the workload scheduler device **305** may select a host device **210**, from the set of host devices **210** within the feasible host set, on which to deploy the requested workload in accordance with the scores or ranks obtained via the scoring **535**. In the example in which the workload scheduler device **305** obtains the first score or rank for the first host device **210** and the second score or rank for the second host device **210**, the workload scheduler device **305** may assign the workload to the first host device **210** or the second host device **210** in accordance with which of the first host device **210** or the second host device **210** has the higher score or rank, the lower score or rank, or the score or rank closer to a target score or rank. For example, in some implementations, if the first score or rank is greater than the second score or rank, the workload scheduler device **305** may assign the workload to the first host device **210**. In some other implementations, if the first score or rank is less than the second score or rank, the workload scheduler device **305** may assign the workload to the first host device **210**.

[0132] FIG. **6** shows an example filtering procedure **600** that supports workload scheduling in edge computing. The filtering procedure **600** may be an example of the filtering **515** as illustrated by and described with reference to FIG. **5**. For example, a workload scheduler device **305** may perform the filtering procedure **600** as part of a workload deployment algorithm, such as the workload deployment algorithm **500** as illustrated and described with reference to FIG. **5**. In some implementations, the workload scheduler device **305** may perform a series or sequence of steps in accordance with (as part of) the filtering procedure **600** to identify, calculate, determine, or ascertain whether a given workload variant is supported by a given host device **210** (in accordance with a capability of the host device **210**). For example, in accordance with the filtering procedure **600**, the workload scheduler device **305** may search over all (or at least a set of) host devices **210** within a distributed computing cluster, add a constraint of power to rule out low power devices, and calculate a score for individual resources when filtering.

[0133] At **605**, the workload scheduler device **305** may receive an input. The input may be associated with a given workload variant and a capability of a given host device **210**. For example, the input may be associated with or otherwise indicate a node data structure (which may be indicative of the capability of the host device **210**) and a workload data structure (which may be indicative of the workload variant).

[0134] At **610**, the workload scheduler device **305** may perform a filtering in accordance with an affinity or selector associated with the workload. If a given host device **210** satisfies a criteria associated with the affinity or selector, the workload scheduler device **305** may pass the host device **210** to a next step of the filtering procedure **600**. Such an affinity or selector associated with the workload may be understood as a host/workload selector/affinity, which may include or refer to any

one or more of a host selector, a workload selector, a workload affinity, or a workload anti-affinity. The host selector (according to which a workload selects a host device **210**) and the workload selector (according to which a host device **210** selects a workload) may be associated with a workload/host dependency (such as a workload/node dependency). The workload affinity and the workload anti-affinity may be associated with a workload/workload dependency.

[0135] In accordance with a host selector, a device (such as a client device **205** or the workload scheduler device **305**) may assign tags to workloads and the workload scheduler device **305** may select host devices **210** (for potential inclusion in the feasible host set) with a matching tag. Such a matching between a tag of a host device **210** and a workload may provide or support a customization of additional constraints in host filtering. For example, in accordance with the host selector, a workload may only select host devices **210** with a dedicated channel connection, host devices **210** that are stationary, or the like.

[0136] In accordance with a workload selector, the workload scheduler device **305** may reserve host devices **210** for workloads with matching tags. For example, the workload scheduler device **305** may reserve a host device **210** with an Ethernet connection (such as with an Ethernet connection to a serving network device) for session-oriented or mission-critical workloads (such as for only session-oriented or mission-critical workloads). Thus, a host device **210** may be understood as selecting a workload in accordance with such a reservation of host devices **210** for workloads with a tag matching a tag of the host device **210**.

[0137] In accordance with a workload affinity, the workload scheduler device **305** may deploy workloads with same tags on a same host device **210**. For example, if two or more workloads are associated with a same tag, the workload scheduler device **305** may assign the two or more workloads to a same host device **210** for processing and workload execution. In such examples, the workload scheduler device **305** may reduce communication costs (if, for example, a single host device **210** may provide spectral efficiency or throughput gains by processing workloads associated with a same tag). For further example, the workload scheduler device **305** may attempt to collocate (such as assign to a single host device **210**) distributed AI workloads (to reduce a communication delay). If the workload scheduler device **305** is unable to collocate distributed AI workloads at a same host device **210**, the workload scheduler device **305** may assign the distributed AI workloads to two or more host devices **210** with a relatively low communication delay among the two or more host devices **210** (such that, for example, the assigned host devices **210** are relatively near each other or otherwise have communication delays between each other that satisfies a threshold communication delay).

[0138] In accordance with a workload anti-affinity, the workload scheduler device **305** may attempt to spread out some workloads across different host devices **210**. For example, the workload scheduler device **305** may attempt to distribute session-oriented workloads on different host devices **210**. In such examples, the workload scheduler device **305** may reduce a communication load of each assigned host device **210** and avoid a scenario in which all (or many, such as several) session-oriented workloads are rescheduled by an assigned host device **210**. Such a rescheduling may occur in examples in which the assigned host device **210** moves (such as exits the distributed computing cluster), launches a relatively higher-priority native application, is subsequently assigned a relatively higher-priority edge compute workload, or suffers an adverse change in link conditions.

[0139] At **615**, if a host device **210** satisfies the criteria associated with the affinity or selector, the workload scheduler device **305** may perform a filtering in accordance with a power constraint associated with the workload. If the host device **210** satisfies a criteria associated with the power constraint, the workload scheduler device **305** may pass the host device **210** to a next step of the filtering procedure **600**. In some examples, the criteria associated with the power constraint may indicate an expectation or constraint for a host device **210** to be grid powered. In such examples, the power constraint criteria may indicate a threshold value `s.sub.threshold` applicable to when the

host device **210** is connected to a power grid. The host device **210** may satisfy the $s_{sub.threshold}$ if $s \geq s_{sub.threshold}$, with s being defined by Equation 1, shown below, and with P being equal to an amount of power provided by the power grid (such as 100,000 Wh). In some aspects, s of Equation 1 may be understood as a score for the host device **210** in terms of power availability.

$$[00001] \ s = \log_{10} P \quad (1)$$

[0140] Additionally, or alternatively, the criteria associated with the power constraint may indicate an expectation or constraint for a host device **210** to have at least a threshold percentage of remaining battery power or at least a threshold amount of remaining battery power. For example, the criteria may include an indication of a threshold percentage of remaining battery power, such as $x_{sub.1}\%$, or an indication of a threshold amount of remaining battery power, such as $x_{sub.2}$ watt hours (Wh).

[0141] At **620**, if a host device **210** satisfies the criteria associated with the power constraint, the workload scheduler device **305** may perform a filtering in accordance with a criteria associated with unquantifiable software or hardware associated with the workload. If the host device **210** satisfies the unquantifiable software or hardware criteria, the workload scheduler device **305** may pass the host device **210** to a next step of the filtering procedure **600**. In some examples, the unquantifiable software or hardware criteria may be associated with or indicate a target OS, a container flag, or a target architecture.

[0142] At **625**, if a host device **210** satisfies the unquantifiable software or hardware criteria, the workload scheduler device **305** may perform a filtering in accordance with a criteria associated with quantifiable hardware associated with the workload. If the host device **210** satisfies the quantifiable hardware criteria, the workload scheduler device **305** may pass the host device **210** to a next step of the filtering procedure **600**. In some examples, the quantifiable hardware criteria may be associated with or indicate an expectation or target amount of computation resources. Such computation resources may include an amount of storage (such as an amount of available storage), an amount of memory (such as an amount of available memory), a quantity of (available) CPU cores, a quantity of (available) GPU cores, or a quantity of (available) NPUs at the host device **210**. If the host device **210** has at least the expected or target amount of computation resources, the host device **210** may pass the filtering at **625**.

[0143] In some examples, a quantifiable hardware criteria may include an indication of one or more $s_{sub.threshold}$ values, each of the one or more $s_{sub.threshold}$ values corresponding to a different threshold (such as lower limit) amount of computation resources. For example, a first $s_{sub.threshold,1}$ value may correspond to a percentage amount of available storage, a second $s_{sub.threshold,2}$ value may correspond to a percentage amount of CPU cores, and so on. In some aspects, each $s_{sub.threshold}$ value may correspond to a lower limit amount of resources left after considering already used resources $r_{sub.used}$ (if applicable) and resources expected to be used to process the workload r , with respect to a total quantity of resources $r_{sub.total}$ at the host device **210**. In such aspects, for example, the host device **210** may satisfy the $s_{sub.threshold,1}$ if $s_{sub.1} \geq s_{sub.threshold,1}$ and may satisfy the $s_{sub.threshold,2}$ if $s_{sub.2} \geq s_{sub.threshold,2}$, with $s_{sub.1}$ and $s_{sub.2}$ being defined by Equations 2 and 3, respectively. Generally, the workload scheduler device **305** may use Equation 2 for memory or storage criteria and may use Equation 3 for CPU, GPU, or NPU criteria. In some aspects, $s_{sub.1}$ and $s_{sub.2}$ of Equations 2 and 3, respectively, may be understood as scores for the host device **210** in terms of computation or processing availability.

$$[00002] \ s_1 = 1 - \frac{r + r_{used}}{r_{total}} \quad (2) \quad s_2 = 1 - \frac{r}{r_{total}} \quad (3)$$

[0144] At **630**, if the host device **210** satisfies the quantifiable hardware criteria, the workload scheduler device **305** may perform a filtering in accordance with an SLA expectation (such as a target QoS) associated with the workload. If the host device **210** satisfies the SLA expectation, the workload scheduler device **305** may pass the host device **210** to a next step of the filtering

procedure **600**. In some implementations, the workload scheduler device **305** may evaluate whether the host device **210** satisfies the SLA expectation if the workload is a session-oriented workload type and may skip the SLA-based evaluation if the workload is not a session-oriented workload type. In other words, the workload scheduler device **305** may evaluate whether the host device **210** satisfies the SLA expectation only if the workload is a session-oriented workload type. In some examples, the SLA expectation may indicate one or more target throughputs or throughput margins, such as one or both of a source-to-host throughput (such as a client-to-host or scheduler-to-host throughput) or a host-to-source throughput (such as a host-to-sink or host-to-client throughput). If the host device **210** satisfies the source-to-host throughput (or throughput margin) and the host-to-source throughput (or throughput margin), the host device **210** may satisfy the SLA expectation. In some aspects, the workload scheduler device **305** may calculate a throughput margin s in accordance with Equation 4, shown below, with $R_{sub.req}$ being a target (such as requested) throughput and with R being an achievable or achieved throughput. In some aspects, s of equation 4 may be understood as a score for the host device **210** in terms of throughput.

$$[00003] \quad s = \frac{R}{R_{req}} \quad (4)$$

[0145] At **635**, if the host device **210** satisfies the relevant criteria (including the SLA expectation if the requested workload is a session-oriented workload type and excluding the SLA expectation if the requested workload is a one-shot workload type), the workload scheduler device **305** may save an ID associated with the host device **210** and a set of one or more scores (such as s values) associated with the host device (with such scores being associated with the relevant criteria). In some aspects, the workload scheduler device **305** may save a score for the host device **210** for a set of metrics associated with one or more of power, memory, storage, quantity of CPU cores, quantity of CPU cores, quantity of NPU cores, or throughput.

[0146] At **640**, in accordance with saving the ID of the host device **210**, the workload scheduler device **305** may add the host device **210** to a feasible host set. Alternatively, if any relevant criteria is not satisfied or met by the host device **210**, the workload scheduler device **305** may exit the path towards adding the host device **210** to the feasible host set and, at **645**, may instead skip to a next host device **210** in a set of potentially feasible host devices **210**. In some implementations, the workload scheduler device **305** may repeat the filtering procedure **600** with the next host device **210**, and each subsequent next host device **210** of the set of potentially feasible host devices **210**, to obtain a feasible host set of one or more host devices **210**.

[0147] FIG. 7 shows an example scoring hierarchy **700** that supports workload scheduling in edge computing. In some examples, a workload scheduler device **305** may leverage or otherwise use the scoring hierarchy **700** to perform the scoring **535** as illustrated by and described with reference to FIG. 5. For example, the workload scheduler device **305** may use the scoring hierarchy **700** to obtain a score **705** for each host device **210** of a set of host devices **210** within a feasible host set. The workload scheduler device **305** may rank the set of host devices **210** of the feasible host set in accordance with the obtained scores **705**, select a host device **210** from the set of host devices **210** in accordance with the ranking, and assign a requested workload to the selected host device **210**.

[0148] In some implementations, the workload scheduler device **305** may obtain a score **705** for a host device **210** in accordance with a joint evaluation, for the host device **210** and in accordance with one or more parameters associated with the workload, of one or more computation metrics **710**, one or more communication metrics **715**, one or more power supply metrics **720**, and one or more reliability metrics **725**. The one or more computation metrics **710** may include one or more CPU metrics **730**, one or more GPU metrics **735**, one or more NPU metrics **740**, one or more memory metrics **745**, or one or more storage metrics **750**. The one or more communication metrics **715** may include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics. The one or more power supply metrics **720** may include one or more power source metrics or one or more battery power metrics.

[0149] In some aspects, a computation metric **710**, a communication metric **715**, and a power supply metric **720** may inform or be associated with a host performance. For example, a computation metric **710** may account for preferences for host devices **210** with relatively more idle (such as available) compute resources (such as CPU, memory, storage, and the like), a communication metric **715** may account for preferences for host devices **210** with stable and relatively high end-to-end throughput, and a power supply metric **720** may account for preferences for host devices **210** with sufficient power supplies. In some aspects, the workload scheduler device **305** may consider a communication metric **715** for session-oriented workload types and may ignore a communication metric **715** for other workload types.

[0150] The one or more reliability metrics **725** may include one or more preemption metrics, one or more mobility metrics **755**, or one or more usage metrics **760**. The one or more mobility metrics **755** may include one or more channel variation metrics **765**, one or more topology change metrics **770**, or one or more cluster exit metrics **775**. The one or more usage metrics **760** may include one or more duration of usage metrics **780** associated with one or more local (such as native) applications, one or more computation resource usage metrics **785** associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0151] In some aspects, a host reliability metric **725** may inform or be associated with a host reliability. For example, the workload scheduler device **305** may use the host reliability metric **725** to account for (such as to reflect) historical usage information that might predict (such as be indicative of) the usage of the host device **210**, historical information that might predict (such as be indicative of) a mobility of the host device **210** (such as occurrences of the host device **210** leaving a cluster, local application preemptions, or communication no longer satisfying expectations), or to evaluate a workload deployment history on the host device **210** (such as to identify, determine, or ascertain how often rescheduling occurs at the host device **210**, such as rescheduling due to local application or higher-priority edge compute workload launches). In accordance with considering the host reliability metric **725**, the workload scheduler device **305** may ascertain that many host devices **210** may not be suitable to provide workload services even if the such host devices **210** have spare computation resources (such as a phone running on low battery, a laptop a user uses often, or an electric vehicle that is driven away).

[0152] In terms of usage history, the workload scheduler device **305** may evaluate one or more metrics or factors associated with how long the host device **210** is accessed each day (such as in terms of hours, such as, for example, four hours) or how much computation the host device **210** performs each day (such as in terms of CPU or memory consumption). In accordance with evaluating such metrics or factors associated with usage history, the workload scheduler device **305** may avoid deploying a workload on a heavily used host device **210** (which may reduce how many preemptions are experienced). In terms of mobility (which may be associated with or informed by the mobility metric **755**), the workload scheduler device **305** may evaluate how often the host device **210** is mobile (such as how often variations in channel conditions occurs, which may relate to a variance of average channel power), how often the host device **210** exits a cluster (which may relate to a quantity or frequency of preemptions), or how long the host device **210** is outside of the cluster. Further, in terms of achieving a balanced workload deployment, the workload scheduler device **305** may attempt to avoid deploying multiple (such as all) session-oriented or mission-critical workloads on one host device **210**, which may protect against scenarios in which a local application or higher-priority edge compute workload is launched at the host device **210** or in which the host device **210** is or becomes mobile.

[0153] In some aspects, the score **705** may be a weighted sum in accordance with the scoring hierarchy **700**. For example, a computation metric **710** (s.sub.compute) may be a weighted sum of the one or more CPU metrics **730**, the one or more GPU metrics **735**, the one or more NPU metrics

740, the one or more memory metrics **745**, and the one or more storage metrics **750**. In some aspects, a first score or value associated with the one or more CPU metrics **730** may be denoted as s.sub.CPU, a second score or value associated with the one or more GPU metrics **735** may be denoted as s.sub.GPU, a third score or value associated with the one or more NPU metrics **740** may be denoted as s.sub.NPU, a fourth score or value associated with the one or more memory metrics **745** may be denoted as s.sub.memory, and a fifth score or value associated with the one or more storage metrics **750** may be denoted as s.sub.storage. In some implementations, the workload scheduler device **305** may apply a respective weighting to each of s.sub.CPU, s.sub.GPU, s.sub.NPU, s.sub.memory, and s.sub.storage. In such implementations, s.sub.compute may be the weighted sum of s.sub.CPU, s.sub.GPU, s.sub.NPU, s.sub.memory, and s.sub.storage (where each of s.sub.CPU, s.sub.GPU, s.sub.NPU, s.sub.memory, and s.sub.storage may be examples of weighed values or where weightings of w.sub.CPU, w.sub.GPU, w.sub.NPU, w.sub.memory, and w.sub.storage may be applied to s.sub.CPU, s.sub.GPU, s.sub.NPU, s.sub.memory, and s.sub.storage, respectively). In other words, the workload scheduler device **305** may calculate a computation metric **710** in accordance with Equation 5, shown below.

$$[00004] S_{\text{compute}} = S_{\text{CPU}} + S_{\text{GPU}} + S_{\text{NPU}} + S_{\text{memory}} + S_{\text{storage}} \quad (5)$$

[0154] For further example, a mobility metric **755** (s.sub.mobility) may be a weighted sum of the one or more channel variation metrics **765**, the one or more topology change metrics **770**, and the one or more cluster exit metrics **775**. In some aspects, a first score or value associated with the one or more channel variation metrics **765** may be denoted as s.sub.channel variations, a second score or value associated with the one or more topology change metrics **770** may be denoted as s.sub.topology change, and a third score or value associated with the one or more cluster exit metrics **775** may be denoted as s.sub.cluster exit. In some implementations, the workload scheduler device **305** may apply a respective weighting to each of s.sub.channel variations, s.sub.topology change, and s.sub.cluster exit. In such implementations, s.sub.mobility may be the weighted sum of s.sub.channel variations, s.sub.topology change, and s.sub.cluster exit (where each of s.sub.channel variations, s.sub.topology change, and s.sub.cluster exit may be examples of weighed values or where weightings of w.sub.channel variations, w.sub.topology change, and w.sub.cluster exit may be applied to s.sub.channel variations, s.sub.topology change, and s.sub.cluster exit, respectively). In other words, the workload scheduler device **305** may calculate a mobility metric **755** in accordance with Equation 6, shown below.

$$[00005] S_{\text{mobility}} = S_{\text{channelvariations}} + S_{\text{topologychange}} + S_{\text{clusterexit}} \quad (6)$$

[0155] For further example, a usage metric **760** (s.sub.usage) may be a weighted sum of the one or more duration of usage metrics **780** and the one or more computation resource usage metrics **785** (potentially among other metrics). In some aspects, a first score or value associated with the one or more duration of usage metrics **780** may be denoted as s.sub.duration of usage and a second score or value associated with the one or more computation resource usage metrics **785** may be denoted as s.sub.amount of computation. In some implementations, the workload scheduler device **305** may apply a respective weighting to each of s.sub.duration of usage and s.sub.amount of computation. In such implementations, s.sub.usage may be the weighted sum of s.sub.duration of usage and s.sub.amount of computation (where each of s.sub.duration of usage and s.sub.amount of computation may be examples of weighed values or where weightings of w.sub.duration of usage and w.sub.amount of computation may be applied to s.sub.duration of usage and s.sub.amount of computation, respectively). In other words, the workload scheduler device **305** may calculate a usage metric **760** in accordance with Equation 7, shown below.

$$[00006] S_{\text{usage}} = S_{\text{durationofusage}} + S_{\text{amountofcomputation}} \quad (7)$$

[0156] In other words, and more generally, the workload scheduler device **305** may calculate the scores of each of the leaf nodes (illustrated in FIG. 7 with dashed-line boxes) of the scoring hierarchy **700** in accordance with scoring functions and may calculate the scores in the non-leaf

nodes (illustrated in FIG. 7 with solid-line boxes) bottom-up, by the weighted sum of its leaf nodes (such as children) such that, for a given non-leaf node, the score s is equal to

$$[00007] s = \frac{\text{Math. } w \times \text{score}}{\text{Math. } w},$$

with w denoting a weighting for a specific score (such that different scores may have different weightings). For example, the workload scheduler device **305** may calculate $s.\text{sub.usage}$ in accordance with Equation 8, shown below, where $w.\text{sub.d}=w.\text{sub.duration}$ of usage, $w.\text{sub.a}=w.\text{sub.amount}$ of computation, $s.\text{sub.d}=s.\text{sub.duration}$ of usage, and $s.\text{sub.a}=s.\text{sub.amount}$ of computation.

$$[00008] s_{\text{usage}} = \frac{(w_d \times s_d) + (w_a \times s_a)}{w_d + w_a} \quad (8)$$

[0157] In some aspects, the workload scheduler device **305** may set all weightings (which may be equivalently referred to herein as weights) to be equal to a default value (such as a one value) in a distributed computing phase **1**, and may adjust the weightings over time based on previous assignments and historical data. Additionally, or alternatively, an application developer or end user may specify the weightings of different resources as part of defining (such as indicating or configuring) the workload inputs.

[0158] In some implementations, the workload scheduler device **305** may normalize the score for each resource (a resource being, for example, a quantifiable hardware resource or computation metric **710**, a communication metric **715**, a power supply metric **720**, or a host reliability metric **725**) given a list of the scores for that resource across the host devices **210** within the feasible host set. For example, given a set of throughput margin scores of $s=[1, 3, 7, 8, 21]$ across a set of five host devices **210** within a feasible host set, the workload scheduler device **305** may normalize the throughput margin scores in accordance with Equation 9, shown below, to obtain updated (such as normalized) scores s' . In accordance with the normalization of scores of each resource, the updated/normalized scores of each resource may be in a range of $s'=[0,1]$. Further, if $s.\text{sub.max}=s.\text{sub.min}$, $s'=1$ for all host devices **210** within the feasible host set.

$$[00009] s' = \frac{s - s_{\min}}{s_{\max} - s_{\min}} \quad (9)$$

[0159] The following relates to an example in which the workload scheduler device **305** calculates, determines, or otherwise obtains scores **705** for three host devices **210** including a first host device **210** (such as a host device **1**), a second host device **210** (such as a host device **2**), and third host device **210** (such as a host device **3**) and in which the workload scheduler device **305** accounts for a computation metric **710**, a communication metric **715**, a power supply metric **720**, and a host reliability metric **725**. In such an example, the workload scheduler device **305** may obtain a first score **705** for the first host device **210**, a second score **705** for the second host device **210**, and a third score **705** for the third host device **210** in accordance with scoring functions across each resource and weighted sums at each non-leaf node of the scoring hierarchy **700**. Table 1, shown below, illustrates an example pre-normalized set of scores for a set of resources associated with the computation metric **710** for each of the three host devices.

TABLE-US-00001 TABLE 1 Computation Resource Scores Pre-Normalization

Host	CPU	GPU	Memory	Storage	NPU	Device Score
Device 1	0.3	0.2	0.9	0.9	0.4	0.8
Device 2	0.4	0.4	0.1	0.8	0.4	0.6
Device 3	0.4	0.1	0.8	0.4	0.1	0.9

[0160] In some examples, the workload scheduler device **305** may normalize the various resource scores with respect to scores across the set of host devices **210**. The workload scheduler device **305** may normalize the scores in accordance with Equation 9, shown above. In accordance with normalizing the various scores, the workload scheduler device **305** may obtain updated (such as normalized) scores as illustrated by Table 2, shown below.

TABLE-US-00002 TABLE 2 Computation Resource Scores Post- Normalization and Pre-Summation

Host	CPU	GPU	Memory	Storage	NPU	Device Score
Device 1	0	0	1	1	0	1
Device 2	0.2	0.5	0	0.875	0	0.875
Device 3	0.875	0	0.875	0	0.875	0.875

[0161] In accordance with normalizing the computation resource scores, the workload scheduler

device 305 may calculate a weighted sum for each host device 210. Such a weighted sum may be the computation metric 710. In accordance with performing the weighted sum, and in examples in which the weightings on the various scores are equal to 1, the workload scheduler device 305 may obtain a first computation metric 710 for the first host device 210 of 0.4, may obtain a second computation metric 710 for the second host device 210 of 0.8, and may obtain a third computation metric 710 for the third host device 210 of 0.32. In accordance with obtaining the computation metrics 710 (such as the computation or compute scores) for the first host device 210, the second host device 210, and the third host device 210, the workload scheduler device 305 may calculate a respective score 705 for each of the three host devices in accordance with factoring in a communication metric 715, a power supply metric 720, and a reliability metric 725. Table 3, shown below, illustrates an example set of scores for the three host devices across computation metrics 710, communication metrics 715, power supply metrics 720, and reliability metrics 725. The metrics may be shown pre-normalization across the different host devices 210.

TABLE-US-00003

TABLE 3 Computation, Communication, Power Supply, and Reliability Scores Pre-Normalization	
Power Host	Computation Communication Supply Reliability Device Score
Score Score	Host
0.4 15 1 3	Device 1
0.8 5 1 6	Device 2
0.32 12 1 9	Device 3

[0162] In some examples, the workload scheduler device 305 may normalize the computation metrics 710, the communication metrics 715, the power supply metrics 720, and the reliability metrics 725 with respect to scores across the set of host devices 210. The workload scheduler device 305 may normalize the metrics or scores in accordance with Equation 9, shown above. In accordance with the normalization, the workload scheduler device 305 may obtain updated (such as normalized) scores as illustrated by Table 4, shown below.

TABLE-US-00004

TABLE 4 Computation, Communication and Power Supply Scores Post-Normalization	
Power Host	Computation Communication Supply Reliability Device Score
Score Score	Host
0.17 1 1 0	Device 1
1 0 1 0.5	Device 2
0 0.7 1 1	Device 3

[0163] In accordance with normalizing the scores, the workload scheduler device 305 may calculate a weighted sum for each host device 210. Such a weighted sum may be the score 705. In accordance with performing the weighted sum, and in examples in which the weightings on the various scores are equal to 1, the workload scheduler device 305 may obtain a first score 705 for the first host device 210 of 0.54, may obtain a second score 705 for the second host device 210 of 0.63, and may obtain a third score 705 for the third host device 210 of 0.68. In such examples, the workload scheduler device 305 may rank the third host device 210 ahead of the second host device 210 and may rank the second host device 210 ahead of the first host device 210 in accordance with their respective scores 705. In accordance with the top ranking of the third host device 210, the workload scheduler device 305 may transmit an indication of an assignment of the workload to at least the third host device 210.

[0164] In some implementations, the workload scheduler device 305 may split a workload across multiple host devices 210 or may duplicate a workload on multiple host devices 210. In such implementations, the workload scheduler device 305 may select multiple host devices 210 with the relatively highest rankings (such as the host devices 210 with the relatively greatest scores, the relatively lowest scores, or the scores relatively closest to a target score) and may split a workload among the selected host devices 210 or duplicate the workload among the selected host devices 210. In other words, the workload scheduler device 305 may transmit multiple indications of multiple assignments of the workload to a set of host devices 210, with such multiple assignments referring to the workload in part or in entirety. In some aspects, the workload scheduler device 305 may segment or split the workload into multiple portions and may assign a portion to a specific host device 210 in accordance with that host device 210 being relatively more suitable (in accordance with one or more metrics associated with that host device 210) for one or more characteristics associated with that portion of the workload. Thus, unless otherwise explicitly indicated, an assignment of a workload to a host device 210 may be understood as an assignment of

a complete workload (such as in its entirety) or as an assignment of a portion of a workload. [0165] In some implementations, the workload scheduler device **305** may have learning capabilities and may weight configuration or host selection through learning that is informed by or otherwise in accordance with past (such as previous) workload logs. With such learning capabilities, the workload scheduler device **305** may define (such as configure or set) the performance metrics for the orchestration or scheduling design or may perform joint configuration (such as joint optimization) of orchestration for an entire system (as opposed to for a single workload). Additional details relating to such learning capabilities are illustrated by and described with reference to FIG. **8**.

[0166] FIG. **8** shows an example scoring adaptation algorithm **800** that supports workload scheduling in edge computing. In some examples, a workload scheduler device **305** may leverage or use the scoring adaptation algorithm **800** to configure, set, update, tune, or adjust one or more weightings associated with a weighted sum-based scoring for each of various feasible host devices **210**, with such a weighted sum-based scoring being an example of the scoring **535** as illustrated by and described with reference to FIG. **5**. For example, the workload scheduler device **305** may perform a joint evaluation for each host device **210** within a feasible host set to obtain weighted sum scores for each host device **210** and may configure, set, update, tune, or adjust the weightings periodically or aperiodically in accordance with system performance, feedback, or user settings.

[0167] For example, the workload scheduler device **305** may perform a host assignment **805** associated with a requested workload in accordance with a first set of weightings. In accordance with the host assignment **805**, the workload scheduler device **305** may obtain, receive, or retrieve a workload log **810** and may perform an analysis **815** of the system performance in association with evaluating the workload log **810**. In some examples, the workload scheduler device **305** may identify a bottleneck of a workload in association with evaluating the workload log **810** and analyzing the system performance. Such a bottleneck may be computation-bound, communication-bound, or the like.

[0168] In some examples, the workload scheduler device **305** may tune (such as configure, set, update, or adjust) one or more weightings of the first set of weightings associated with the weighted sum scoring in accordance with the workload log **810** (or multiple workload logs **810**), a workload type, or workload expectations. In such examples, the workload scheduler device **305** may tune the one or more weightings in accordance with one or both of domain knowledge **825** or feedback **830**. Additionally, or alternatively, the workload scheduler device **305** may treat a box **820** (including the domain knowledge **825** and the feedback **830**) as a block box and the workload scheduler device **305** may attempt to learn the box **820** via a learning capability (such as a learning algorithm) of the workload scheduler device **305**. In other words, the workload scheduler device **305** may use the box **820** as a weighting configuration or may treat the scoring as a block box. In accordance with the adaptation algorithm **800**, the workload scheduler device **305** may update the first set of weightings to a second set of weightings and may use the second set of weightings for subsequent scoring (such as for subsequent joint evaluations associated with subsequent workload requests).

[0169] FIG. **9** shows an example process flow **900** that supports workload scheduling in edge computing. The process flow **900** may implement or be implemented to realize one or more aspects of the wireless communication system **100**, the connected edge cluster architecture **200**, the signaling diagram **300**, the workload scheduling diagram **400**, the nested workload scheduling algorithm **401**, the workload deployment algorithm **500**, the filtering procedure **600**, the scoring hierarchy **700**, and the adaptation algorithm **800**. For example, the process flow **900** illustrates communication between a client devices **205**, a workload scheduler device **305**, and a host device **210**.

[0170] Alternative examples of the following may be implemented. Some steps are performed in a different order than described or are not performed at all. In some implementations, steps may

include additional features not mentioned below, or further steps may be added. Further, although example devices are shown performing the operations of the process flow **900**, some aspects of some operations also may be performed by one or more other wireless communication devices without exceeding the scope of the present disclosure.

[0171] At **905**, the workload scheduler device **305** may receive, from the host device **210**, information indicative of a set of one or more computation metrics associated with the host device **210**, a set of one or more communication metrics associated with the host device **210**, a set of one or more power supply metrics associated with the host device **210**, and a set of one or more reliability metrics associated with the host device **210**. Further, in some implementations, the workload scheduler device **305** may receive respective information of such metrics from each host device **210** of a set of host devices **210**. In such implementations, the workload scheduler device **305** may receive, from each host device **210**, information indicative of a respective set of one or more computation metrics associated with that host device **210**, a respective set of one or more communication metrics associated with that host device **210**, a respective set of one or more power supply metrics associated with that host device **210**, and a respective set of one or more reliability metrics associated with that host device **210**.

[0172] At **910**, the workload scheduler device **305** may receive, from the client device **205**, a workload request. For example, the workload scheduler device **305** may receive a request associated with a workload. In some implementations, the workload scheduler device **305** may receive, via the workload request or via other signaling or messaging, information indicative of one or more parameters associated with the workload.

[0173] At **915**, the workload scheduler device **305** may perform a filtering, such as a filtering **515** as illustrated by and described with reference to FIG. 5. Further, in some aspects, such filtering may include a series or sequence of steps of a filtering procedure, such as the filtering procedure **600** as illustrated by and described with reference to FIG. 6. In accordance with the filtering, the workload scheduler device **305** may obtain identifying information for (such as IDs of) a set of host devices **210** from a larger set of potential host devices in association with evaluating each host device **210** of the larger set of host devices **210**. The workload scheduler device **305** may evaluate each host device **210** in accordance with the parameter(s) associated with the workload and one or more variants of the workload. Such a set of host devices **210** may be understood as a feasible host set, and may include one or multiple members. Some example variants of the workload may include a workload variant **510-a**, a workload variant **510-b**, and a workload variant **510-c** as illustrated by and described with reference to FIG. 5.

[0174] At **920**, the workload scheduler device **305** may perform a scoring, such as a scoring **535** as illustrated by and described with reference to FIG. 5. Further, in some aspects, such scoring may be in accordance with a scoring hierarchy, such as the scoring hierarchy **700** as illustrated by and described with reference to FIG. 7. In accordance with the scoring, the workload scheduler device **305** may obtain a set of scores (such as a set of scores **705**) for the set of host devices **210** (such as for host devices **210** within the feasible host set) in accordance with a respective joint evaluation, for each host device **210** of the set of host devices **210** and in accordance with the parameter(s) associated with the workload, of one or more computation metrics, one or more communication metrics, one or more power supply metrics, and one or more reliability metrics. Such metrics may include the one or more computation metrics **710**, the one or more communication metrics **715**, the one or more power supply metrics **720**, and the one or more host reliability metrics **725** as illustrated by and described with reference to FIG. 7.

[0175] At **925**, in accordance with the scoring, the workload scheduler device **305** may transmit an indication of an assignment of the workload to at least the host device **210**. In some implementations, the workload scheduler device **305** may assign the workload to the host device **210** in accordance with a first score for the host device **210** being relatively greater than one or more scores for a remainder of the set of host devices **210**.

[0176] At **930**, the workload scheduler device **305** may transmit, to the host device **210** or another edge device (such as a network device, such as an AP, gNB, or router, serving the host device **210**), link configuration information. In some implementations, the workload scheduler device **305** may transmit the link configuration information in association with receiving, from the host device **210**, information indicating that the host device **210** is processing connected edge workloads (such as distributed computing workloads). Additionally, or alternatively, a network device (such as a Wi-Fi AP or a cellular gNB) may transmit the link configuration information to the host device **210**. The link configuration information may include an indication to move or steer the host device **210** to a different channel, to a different RF band, or to a different network device. For example, the link configuration information may steer the host device **210** to a first network device from a second network device. In such examples, the workload scheduler device **305** may steer the host device **210** to the first network device in accordance with the first network device being a relatively more reliable or efficient network device in terms of supporting the connected edge workloads at the host device **210**.

[0177] At **935**, the host device **210** may transmit, to the workload scheduler device **305**, an indication of a rejection of the assignment of the workload to the host device **210**. In some implementations, the host device **210** may transmit the indication of the rejection in accordance with one or more percentage metrics associated with an amount of resources used for connected edge workloads, including the assigned workload, at the host device **210** exceeding one or more threshold percentages. Such one or more threshold percentages may include the third threshold percentage $x.sub.3\%$ as described with reference to FIG. 3.

[0178] At **940**, the host device **210**, if the host device **210** accepts the assigned workload (which the host device **210** may signal to the workload scheduler device **305** or which may be implied in the absence of a rejection at **935**), may perform a workload processing. In other words, the host device **210** may perform one or more processing operations or tasks associated with the assigned workload.

[0179] At **945**, the host device **210** may transmit, to the client device **205**, information associated with the workload in accordance with the assignment of the workload to at least the host device **210** (and in accordance with processing the workload at **940**). Such information may include one or more outputs or a result associated with the workload, such as one or more rendered images, one or more computation results, processed visual or audio data, or the like.

[0180] At **950**, the workload scheduler device **305** may receive, from one or both of the client device **205** or the host device **210**, information indicative of one or more network congestion metrics. In some aspects, such one or more network congestion metrics may be associated with a cluster (such as a distributed computing cluster) of edge devices including the feasible host set.

[0181] At **955**, the host device **210** may transmit, to the workload scheduler device **305**, information indicative of a preemption rate associated with workloads assigned to the host device **210**. Additionally, in some implementations, the workload scheduler device **305** may receive similar information for multiple host devices **210** indicating that a preemption rate associated with workloads assigned to a cluster of edge devices.

[0182] At **960**, if the network congestion metric(s) indicate that the network is congested or if the preemption rate exceeds (such as fails to satisfy) a threshold preemption rate, the workload scheduler device **305** may transmit an indication of a suspension of workload requests. Such a suspension of workload requests may be to suspend or reject workloads of a specific type or may be to suspend or reject workloads of any type. For example, if the network congestion metric(s) indicate that the network is congested, the workload scheduler device **305** may suspend or reject workloads associated with a session-oriented workload type. Additionally, or alternatively, if the preemption rate exceeds the threshold preemption rate, the workload scheduler device **305** may suspend or reject workloads associated with any workload type.

[0183] FIG. 10 shows a block diagram of an example device **1005** that supports workload

scheduling in edge computing. The device **1005** may communicate with one or more workload scheduling devices, one or more host devices, one or more client devices, one or more network entities **105**, one or more UEs **115**, or any combination thereof, which may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device **1005** may include components that support outputting and obtaining communications, such as a communication manager **1020**, a transceiver **1010**, one or more antennas **1015**, at least one memory **1025**, code **1030**, and at least one processor **1035**. These components may be in electronic communication or otherwise coupled (such as operatively, communicatively, functionally, electronically, electrically) via one or more buses (such as a bus **1040**).

[0184] The transceiver **1010** may support bi-directional communications via wired links, wireless links, or both as described herein. In some implementations, the transceiver **1010** may include a wired transceiver and may communicate bi-directionally with another wired transceiver. Additionally, or alternatively, in some implementations, the transceiver **1010** may include a wireless transceiver and may communicate bi-directionally with another wireless transceiver. In some implementations, the device **1005** may include one or more antennas **1015**, which may be capable of transmitting or receiving wireless transmissions (such as concurrently). The transceiver **1010** also may include a modem to modulate signals, to provide the modulated signals for transmission (such as by one or more antennas **1015** or by a wired transmitter), to receive modulated signals (such as from one or more antennas **1015** or from a wired receiver), and to demodulate signals. In some implementations, the transceiver **1010** may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas **1015** that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas **1015** that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver **1010** may include or be configured for coupling with one or more processors or one or more memory components that are operable to perform or support operations in accordance with received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver **1010**, or the transceiver **1010** and the one or more antennas **1015**, or the transceiver **1010** and the one or more antennas **1015** and one or more processors or one or more memory components (such as the at least one processor **1035**, the at least one memory **1025**, or both), may be included in a chip or chip assembly that is installed in the device **1005**. In some implementations, the transceiver **1010** may be operable to support communications via one or more communications links (such as communication link(s) **125**, backhaul communication link(s) **120**, a midhaul communication link **162**, a fronthaul communication link **168**).

[0185] The at least one memory **1025** may include random-access memory (RAM), read-only memory (ROM), or any combination thereof. The at least one memory **1025** may store computer-readable, computer-executable, or processor-executable code, such as the code **1030**. The code **1030** may include instructions that, when executed by one or more of the at least one processor **1035**, cause the device **1005** to perform various functions described herein. The code **1030** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some implementations, the code **1030** may not be directly executable by a processor of the at least one processor **1035** but may cause a computer (such as when compiled and executed) to perform functions described herein. In some implementations, the at least one memory **1025** may include, among other things, a basic input/output (I/O) system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some implementations, the at least one processor **1035** may include multiple processors and the at least one memory **1025** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories which may, individually or collectively,

be configured to perform various functions described herein (such as, part of a processing system, a memory system, or both).

[0186] The at least one processor **1035** may include one or more intelligent hardware devices (such as one or more general-purpose processors, one or more digital signal processors (DSPs), one or more CPUs, one or more GPUs, one or more NPUs (also referred to as neural network processors or deep learning processors (DLPs)), one or more microcontrollers, one or more application-specific integrated circuits (ASICs), one or more field-programmable gate arrays (FPGAs), one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some implementations, the at least one processor **1035** may be configured to operate a memory array using a memory controller. In some other implementations, a memory controller may be integrated into one or more of the at least one processor **1035**. The at least one processor **1035** may be configured to execute computer-readable instructions stored in a memory (such as one or more of the at least one memory **1025**) to cause the device **1005** to perform various functions (such as functions or tasks supporting workload scheduling in edge computing). For example, the device **1005** or a component of the device **1005** may include at least one processor **1035** and at least one memory **1025** coupled with one or more of the at least one processor **1035**, the at least one processor **1035** and the at least one memory **1025** configured to perform various functions described herein. The at least one processor **1035** may be an example of a cloud-computing platform (such as one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (such as by executing code **1030**) to perform the functions of the device **1005**. The at least one processor **1035** may be any one or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the device **1005** (such as within one or more of the at least one memory **1025**). In some implementations, the at least one processor **1035** may include multiple processors and the at least one memory **1025** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein.

[0187] In some implementations, the at least one processor **1035** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **1035**) and memory circuitry (which may include the at least one memory **1025**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **1035** or a processing system including the at least one processor **1035** may be configured to, configurable to, or operable to cause the device **1005** to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code stored in the at least one memory **1025** or otherwise, to perform one or more of the functions described herein.

[0188] In some implementations, a processing system of the device **1005** may refer to a system including the various other components or subcomponents of the device **1005**, such as the at least one processor **1035**, or the transceiver **1010**, or the communication manager **1020**, or other components or combinations of components of the device **1005**. The processing system of the device **1005** may interface with other components of the device **1005**, and may process information received from other components (such as inputs or signals) or output information to other components. For example, a chip or modem of the device **1005** may include a processing system and one or more interfaces to output information, or to obtain information, or both.

[0189] The one or more interfaces may be implemented as or otherwise include a first interface configured to output information and a second interface configured to obtain information, or a same interface configured to output information and to obtain information, among other

implementations. In some implementations, the one or more interfaces may refer to an interface between the processing system of the chip or modem and a transmitter, such that the device **1005** may transmit information output from the chip or modem. Additionally, or alternatively, in some implementations, the one or more interfaces may refer to an interface between the processing system of the chip or modem and a receiver, such that the device **1005** may obtain information or signal inputs, and the information may be passed to the processing system. A person having ordinary skill in the art will readily recognize that a first interface also may obtain information or signal inputs, and a second interface also may output information or signal outputs.

[0190] The device **1005** may include one or more chips, SoCs, chipsets, packages or devices that individually or collectively constitute or include a processing system. The processing system includes processor (or “processing”) circuitry in the form of one or multiple processors, microprocessors, processing units (such as CPUs, GPUs, NPUs, or DSPs), processing blocks, ASIC, programmable logic devices (such as FPGAs), or other discrete gate or transistor logic or circuitry (all of which may be generally referred to herein individually as “processors” or collectively as “the processor” or “the processor circuitry”). One or more of the processors may be individually or collectively configurable or configured to perform various functions or operations described herein. The processing system may further include memory circuitry in the form of one or more memory devices, memory blocks, memory elements or other discrete gate or transistor logic or circuitry, each of which may include tangible storage media such as RAM or ROM, or combinations thereof (all of which may be generally referred to herein individually as “memories” or collectively as “the memory” or “the memory circuitry”). One or more of the memories may be coupled with one or more of the processors and may individually or collectively store processor-executable code that, when executed by one or more of the processors, may configure one or more of the processors to perform various functions or operations described herein. Additionally, or alternatively, in some implementations, one or more of the processors may be preconfigured to perform various functions or operations described herein without requiring configuration by software. The processing system may further include or be coupled with one or more modems (such as a Wi-Fi (such as IEEE compliant) modem or a cellular (such as 3GPP 4G LTE, 5G, or 6G compliant) modem). In some implementations, one or more processors of the processing system include or implement one or more of the modems. The processing system may further include or be coupled with multiple radios (collectively “the radio”), multiple RF chains, or multiple transceivers, each of which may in turn be coupled with one or more of multiple antennas. In some implementations, one or more processors of the processing system include or implement one or more of the radios, RF chains, or transceivers.

[0191] In some implementations, a bus **1040** may support communications of (such as within) a protocol layer of a protocol stack. In some implementations, a bus **1040** may support communications associated with a logical channel of a protocol stack (such as between protocol layers of a protocol stack), which may include communications performed within a component of the device **1005**, or between different components of the device **1005** that may be co-located or located in different locations (such as where the device **1005** may refer to a system in which one or more of the communication manager **1020**, the transceiver **1010**, the at least one memory **1025**, the code **1030**, and the at least one processor **1035** may be located in one of the different components or divided between different components).

[0192] In some implementations, the communication manager **1020** may manage aspects of communications with a core network **130** (such as via one or more wired or wireless backhaul links). For example, the communication manager **1020** may manage the transfer of data communications for client devices, such as one or more UEs **115**. In some implementations, the communication manager **1020** may manage communications with one or more other network entities **105**, and may include a controller or scheduler for controlling communications with UEs **115** (such as in cooperation with the one or more other network devices). In some implementations,

the communication manager **1020** may support an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between network entities **105**. [0193] In some implementations, the communication manager **1020** (and, likewise, the device **1005**) may support connected edge workload scheduling as or as a component of a workload scheduler device in accordance with examples as disclosed herein. The communication manager **1020** is capable of, configured to, or operable to support a means for receiving, via a request associated with a workload, information indicative of one or more parameters associated with the workload. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0194] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving, from the first host device, information associated with the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0195] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for obtaining a set of scores for a set of host devices including the first host device in accordance with a respective joint evaluation associated with each host device of a set of host devices, where each score of the set of scores is for a respective host device of the set of host devices, and where the assignment of the workload to at least the first host device is in accordance with a first score for the first host device being relatively greater than one or more scores for a remainder of the set of host devices excluding the first host device.

[0196] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving, from each host device of the set of host devices, information associated with a respective set of one or more computation metrics associated with that host device, a respective set of one or more communication metrics associated with that host device, a respective set of one or more power supply metrics associated with that host device, and a respective set of one or more reliability metrics associated with that host device, where the respective joint evaluation associated with each host device of the set of host devices is in accordance with the information received from each host device of the set of host devices.

[0197] In some examples, to support obtaining the set of scores for the set of host devices, the communication manager **1020** is capable of, configured to, or operable to support a means for obtaining the first score for the first host device in accordance with the first joint evaluation. In some examples, to support obtaining the set of scores for the set of host devices, the communication manager **1020** is capable of, configured to, or operable to support a means for obtaining a second score for a second host device of the set of host devices in accordance with a second joint evaluation, associated with the one or more parameters, of a second set of one or more computation metrics associated with the second host device, a second set of one or more communication metrics associated with the second host device, a second set of one or more power supply metrics associated with the second host device, and a second set of one or more reliability metrics associated with the second host device, where the first score is relatively greater than the second score.

[0198] In some examples, each score of the set of scores is a respective weighted sum of a respective set of one or more computation metrics, a respective set of one or more communication metrics, a respective set of one or more power supply metrics, and a respective set of one or more reliability metrics.

[0199] In some examples, to obtain the respective weighted sum, a first weighting is applied to the respective set of one or more computation metrics, a second weighting is applied to the respective set of one or more communication metrics, a third weighting is applied to the respective set of one or more power supply metrics, and a fourth weighting is applied to the respective set of one or more reliability metrics.

[0200] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for obtaining an indication of the first weighting, the second weighting, the third weighting, and the fourth weighting in accordance with historical data associated with workload scheduling across the set of host devices or the one or more parameters associated with the workload.

[0201] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for obtaining a set of host devices from a set of multiple host devices in association with evaluating each host device of the set of multiple host devices in accordance with the one or more parameters and one or more variants of the workload, where the set of host devices includes the first host device.

[0202] In some examples, to support evaluating each host device of the set of multiple host devices, the communication manager **1020** is capable of, configured to, or operable to support a means for evaluating whether the workload is feasible for a respective host device of the set of multiple host devices in accordance with the one or more parameters, a respective set of one or more variants of the workload, and a capability of the respective host device.

[0203] In some examples, the capability of the respective host device is associated with one or more percentage metrics associated with an amount of resources used for one or more existing edge compute workloads at the respective host device, one or more memory usage metrics, one or more storage usage metrics, one or more power metrics at the respective host device, a software or hardware capability of the respective host device, or one or more communication metrics associated with over-the-air signaling between the respective host device and a client device associated with the workload.

[0204] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for including the first host device in the set of host devices in accordance with the workload being feasible for the first host device, where the workload is feasible for the first host device in accordance with the one or more parameters, a first set of one or more variants of the workload, and a first capability of the first host device. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for excluding a second host device of the set of multiple host devices from the set of host devices in accordance with the workload being unfeasible for the second host device, where the workload is unfeasible for the second host device in accordance with the one or more parameters, a second set of one or more variants of the workload, and a second capability of the first host device.

[0205] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving, from the first host device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0206] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting an indication of an updated assignment of the workload to at least a second host device in association with receiving the rejection of the assignment of the workload to the first host device.

[0207] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device. In some examples, the communication manager **1020** is capable of,

configured to, or operable to support a means for transmitting, to the first host device or a first network device associated with the first host device, link configuration information that steers the first host device to the first network device from a second network device in association with the first host device processing the edge compute workloads.

[0208] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with the one or more network congestion metrics.

[0209] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving information indicating that a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeds a threshold preemption rate. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting an indication of a suspension of workload requests in accordance with the preemption rate exceeding the threshold preemption rate.

[0210] In some examples, the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0211] In some examples, the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0212] In some examples, the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0213] Additionally, or alternatively, the communication manager **1020** (and, likewise, the device **1005**) may support connected edge workload scheduling as or as a component of a host device in accordance with examples as disclosed herein. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics,

and the first set of one or more reliability metrics.

[0214] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting, to a client device requesting the workload, information associated with the workload in accordance with the assignment of the workload to at least the first host device.

[0215] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting, to the workload scheduler device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0216] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting, to the workload scheduler device, information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device. In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving link configuration information that steers the first host device to a first network device from a second network device in association with the first host device processing the edge compute workloads.

[0217] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving, from the workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0218] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting, to the workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0219] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for receiving, from the workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0220] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for transmitting, to the workload scheduler device, information indicating that a preemption rate at the first host device exceeds a threshold preemption rate.

[0221] In some examples, the communication manager **1020** is capable of, configured to, or operable to support a means for preempting one or more workloads assigned to the first host device in accordance with one or more network changes, cluster exits, higher-priority edge compute workload launches, or local application launches, where transmitting the information indicating that the preemption rate at the first host device exceeds the threshold preemption rate is in association with preempting the one or more workloads.

[0222] In some examples, the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0223] In some examples, the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute

workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage metrics associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0224] In some examples, the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0225] In some examples, the communication manager **1020** may be configured to perform various operations (such as receiving, obtaining, monitoring, outputting, transmitting) using or otherwise in cooperation with the transceiver **1010**, the one or more antennas **1015** (such as where applicable), or any combination thereof. Although the communication manager **1020** is illustrated as a separate component, in some examples, one or more functions described with reference to the communication manager **1020** may be supported by or performed by the transceiver **1010**, one or more of the at least one processor **1035**, one or more of the at least one memory **1025**, the code **1030**, or any combination thereof (such as by a processing system including at least a portion of the at least one processor **1035**, the at least one memory **1025**, the code **1030**, or any combination thereof). For example, the code **1030** may include instructions executable by one or more of the at least one processor **1035** to cause the device **1005** to perform various aspects of workload scheduling in edge computing as described herein, or the at least one processor **1035** and the at least one memory **1025** may be otherwise configured to, individually or collectively, perform or support such operations.

[0226] FIG. **11** shows a block diagram of an example device **1105** that supports workload scheduling in edge computing. The device **1105** may communicate with one or more workload scheduling devices, one or more host devices, one or more network entities **105**, one or more UEs **115**, or any combination thereof, which may include communications over one or more wired interfaces, over one or more wireless interfaces, or any combination thereof. The device **1105** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, such as a communication manager **1120**, an input/output (I/O) controller, such as an I/O controller **1110**, a transceiver **1115**, one or more antennas **1125**, at least one memory **1130**, code **1135**, and at least one processor **1140**. These components may be in electronic communication or otherwise coupled (such as operatively, communicatively, functionally, electronically, electrically) via one or more buses (such as a bus **1145**).

[0227] The I/O controller **1110** may manage input and output signals for the device **1105**. The I/O controller **1110** also may manage peripherals not integrated into the device **1105**. In some implementations, the I/O controller **1110** may represent a physical connection or port to an external peripheral. In some implementations, the I/O controller **1110** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally, or alternatively, the I/O controller **1110** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some implementations, the I/O controller **1110** may be implemented as part of a processor or processing system, such as the processor **1140**. In some implementations, a user may interact with the device **1105** via the I/O controller **1110** or via hardware components controlled by the I/O controller **1110**.

[0228] In some implementations, the device **1105** may include a single antenna. However, in some other implementations, the device **1105** may have more than one antenna, which may be capable of concurrently transmitting or receiving multiple wireless transmissions. The transceiver **1115** may

communicate bi-directionally via the one or more antennas **1125** using wired or wireless links as described herein. For example, the transceiver **1115** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1115** also may include a modem to modulate the packets, to provide the modulated packets to one or more antennas **1125** for transmission, and to demodulate packets received from the one or more antennas **1125**. In some implementations, the transceiver **1115** may include one or more interfaces, such as one or more interfaces coupled with the one or more antennas **1125** that are configured to support various receiving or obtaining operations, or one or more interfaces coupled with the one or more antennas **1125** that are configured to support various transmitting or outputting operations, or a combination thereof. In some implementations, the transceiver **1115** may include or be configured for coupling with one or more processors or memory components that are operable to perform or support operations based on received or obtained information or signals, or to generate information or other signals for transmission or other outputting, or any combination thereof. In some implementations, the transceiver **1115**, or the transceiver **1115** and the one or more antennas **1125**, or the transceiver **1115** and the one or more antennas **1125** and one or more processors or memory components (such as the processor **1140**, or the memory **1130**, or both), may be included in a chip or chip assembly that is installed in the device **1105**.

[0229] The at least one memory **1130** may include RAM, ROM, or any combination thereof. The at least one memory **1130** may store computer-readable, computer-executable, or processor-executable code, such as code **1135**. The code **1135** may include instructions that, when executed by one or more of the at least one processor **1140**, cause the device **1105** to perform various functions described herein. The code **1135** may be stored in a non-transitory computer-readable medium such as system memory or another type of memory. In some implementations, the code **1135** may not be directly executable by a processor of the at least one processor **1140** but may cause a computer (such as when compiled and executed) to perform functions described herein. In some implementations, the at least one memory **1130** may include, among other things, a basic I/O system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some implementations, the at least one processor **1140** may include multiple processors and the at least one memory **1130** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories which may, individually or collectively, be configured to perform various functions herein (such as part of a processing system, a memory system, or both).

[0230] The at least one processor **1140** may include one or more intelligent hardware devices (such as one or more general-purpose processors, one or more DSPs, one or more CPUs, one or more GPUs, one or more NPU's (also referred to as neural network processors or DLPs), one or more microcontrollers, one or more ASICs, one or more FPGAs, one or more programmable logic devices, discrete gate or transistor logic, one or more discrete hardware components, or any combination thereof). In some examples, the at least one processor **1140** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into one or more of the at least one processor **1140**. The at least one processor **1140** may be configured to execute computer-readable instructions stored in a memory (such as one or more of the at least one memory **1130**) to cause the device **1105** to perform various functions (such as functions or tasks supporting workload scheduling in edge computing). For example, the device **1105** or a component of the device **1105** may include at least one processor **1140** and at least one memory **1130** coupled with one or more of the at least one processor **1140**, the at least one processor **1140** and the at least one memory **1130** configured to perform various functions described herein. The at least one processor **1140** may be an example of a cloud-computing platform (such as one or more physical nodes and supporting software such as operating systems, virtual machines, or container instances) that may host the functions (such as by executing code **1135**) to perform the functions of the device **1105**. The at least one processor **1140** may be any one

or more suitable processors capable of executing scripts or instructions of one or more software programs stored in the device **1105** (such as within one or more of the at least one memory **1130**). In some implementations, the at least one processor **1140** may include multiple processors and the at least one memory **1130** may include multiple memories. One or more of the multiple processors may be coupled with one or more of the multiple memories, which may, individually or collectively, be configured to perform various functions herein.

[0231] In some implementations, the at least one processor **1140** may be a component of a processing system, which may refer to a system (such as a series) of machines, circuitry (including, for example, one or both of processor circuitry (which may include the at least one processor **1140**) and memory circuitry (which may include the at least one memory **1130**)), or components, that receives or obtains inputs and processes the inputs to produce, generate, or obtain a set of outputs. The processing system may be configured to perform one or more of the functions described herein. For example, the at least one processor **1140** or a processing system including the at least one processor **1140** may be configured to, configurable to, or operable to cause the device **1105** to perform one or more of the functions described herein. Further, as described herein, being “configured to,” being “configurable to,” and being “operable to” may be used interchangeably and may be associated with a capability, when executing code stored in the at least one memory **1130** or otherwise, to perform one or more of the functions described herein.

[0232] In some implementations, a processing system of the device **1105** may refer to a system including the various other components or subcomponents of the device **1105**, such as the at least one processor **1140**, or the transceiver **1115**, or the communication manager **1120**, or other components or combinations of components of the device **1105**. The processing system of the device **1105** may interface with other components of the device **1105**, and may process information received from other components (such as inputs or signals) or output information to other components. For example, a chip or modem of the device **1105** may include a processing system and one or more interfaces to output information, or to obtain information, or both.

[0233] The one or more interfaces may be implemented as or otherwise include a first interface configured to output information and a second interface configured to obtain information, or a same interface configured to output information and to obtain information, among other implementations. In some implementations, the one or more interfaces may refer to an interface between the processing system of the chip or modem and a transmitter, such that the device **1105** may transmit information output from the chip or modem. Additionally, or alternatively, in some implementations, the one or more interfaces may refer to an interface between the processing system of the chip or modem and a receiver, such that the device **1105** may obtain information or signal inputs, and the information may be passed to the processing system. A person having ordinary skill in the art will readily recognize that a first interface also may obtain information or signal inputs, and a second interface also may output information or signal outputs.

[0234] The device **1105** may include one or more chips, SoCs, chipsets, packages or devices that individually or collectively constitute or include a processing system. The processing system includes processor (or “processing”) circuitry in the form of one or multiple processors, microprocessors, processing units (such as CPUs, GPUs, NPUs, or DSPs), processing blocks, ASICs, programmable logic devices (such as FPGAs), or other discrete gate or transistor logic or circuitry (all of which may be generally referred to herein individually as “processors” or collectively as “the processor” or “the processor circuitry”). One or more of the processors may be individually or collectively configurable or configured to perform various functions or operations described herein. The processing system may further include memory circuitry in the form of one or more memory devices, memory blocks, memory elements or other discrete gate or transistor logic or circuitry, each of which may include tangible storage media such as RAM or ROM, or combinations thereof (all of which may be generally referred to herein individually as “memories” or collectively as “the memory” or “the memory circuitry”). One or more of the memories may be

coupled with one or more of the processors and may individually or collectively store processor-executable code that, when executed by one or more of the processors, may configure one or more of the processors to perform various functions or operations described herein. Additionally, or alternatively, in some implementations, one or more of the processors may be preconfigured to perform various functions or operations described herein without requiring configuration by software. The processing system may further include or be coupled with one or more modems (such as a Wi-Fi (such as IEEE compliant) modem or a cellular (such as 3GPP 4G LTE, 5G, or 6G compliant) modem). In some implementations, one or more processors of the processing system include or implement one or more of the modems. The processing system may further include or be coupled with multiple radios (collectively “the radio”), multiple RF chains, or multiple transceivers, each of which may in turn be coupled with one or more of multiple antennas. In some implementations, one or more processors of the processing system include or implement one or more of the radios, RF chains, or transceivers.

[0235] In some implementations, a bus **1145** may support communications of (such as within) a protocol layer of a protocol stack. In some implementations, a bus **1145** may support communications associated with a logical channel of a protocol stack (such as between protocol layers of a protocol stack), which may include communications performed within a component of the device **1105**, or between different components of the device **1105** that may be co-located or located in different locations (such as where the device **1105** may refer to a system in which one or more of the communication manager **1120**, the transceiver **1115**, the at least one memory **1130**, the code **1135**, and the at least one processor **1140** may be located in one of the different components or divided between different components).

[0236] The communication manager **1120** (and, likewise, the device **1105**) may support connected edge workload scheduling as or as a component of a client device in accordance with examples as disclosed herein. The communication manager **1120** is capable of, configured to, or operable to support a means for transmitting, via a request associated with a workload, information indicative of one or more parameters associated with the workload. In some examples, the communication manager **1120** is capable of, configured to, or operable to support a means for receiving information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0237] In some examples, the communication manager **1120** is capable of, configured to, or operable to support a means for receiving, from a workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0238] In some examples, the communication manager **1120** is capable of, configured to, or operable to support a means for transmitting, to a workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0239] In some examples, the communication manager **1120** is capable of, configured to, or operable to support a means for receiving, from a workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0240] In some examples, the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or

more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0241] In some examples, the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage metrics associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0242] In some examples, the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0243] In some implementations, the communication manager **1120** may be configured to perform various operations (such as receiving, monitoring, transmitting) using or otherwise in cooperation with the transceiver **1115**, the one or more antennas **1125**, or any combination thereof. Although the communication manager **1120** is illustrated as a component of the transceiver **1115**, in some implementations, one or more functions described with reference to the communication manager **1120** may be supported by or performed by the transceiver **1115**, the at least one processor **1140**, the at least one memory **1130**, the code **1135**, or any combination thereof (such as by a processing system including at least a portion of the at least one processor **1140**, the at least one memory **1130**, the code **1135**, or any combination thereof). For example, the code **1135** may include instructions executable by one or more of the at least one processor **1140** to cause the device **1105** to perform various aspects of workload scheduling in edge computing as described herein, or the at least one processor **1140** and the at least one memory **1130** may be otherwise configured to perform or support such operations.

[0244] FIG. **12** shows a flowchart illustrating an example method **1200** that supports workload scheduling in edge computing. The operations of the method **1200** may be implemented by a workload scheduler device or its components as described herein. For example, the operations of the method **1200** may be performed by a workload scheduler device or a network entity **105** as described with reference to FIGS. **1-10**. In some examples, a workload scheduler device may execute a set of instructions to control the functional elements of the workload scheduler device to perform the described functions. Additionally, or alternatively, the workload scheduler device may perform aspects of the described functions using special-purpose hardware.

[0245] At **1205**, the method may include receiving, via a request associated with a workload, information indicative of one or more parameters associated with the workload. The operations of **1205** may be performed in accordance with examples as disclosed herein.

[0246] At **1210**, the method may include transmitting an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device. The operations of **1210** may be performed in accordance with examples as disclosed herein.

[0247] FIG. **13** shows a flowchart illustrating an example method **1300** that supports workload

scheduling in edge computing. The operations of the method **1300** may be implemented by a first host device or its components as described herein. For example, the operations of the method **1300** may be performed by a first host device or a network entity **105** as described with reference to FIGS. **1-10**. In some examples, a first host device may execute a set of instructions to control the functional elements of the host device to perform the described functions. Additionally, or alternatively, the first host device may perform aspects of the described functions using special-purpose hardware.

[0248] At **1305**, the method may include transmitting, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device. The operations of **1305** may be performed in accordance with examples as disclosed herein.

[0249] At **1310**, the method may include receiving, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics. The operations of **1310** may be performed in accordance with examples as disclosed herein.

[0250] FIG. **14** shows a flowchart illustrating an example method **1400** that supports workload scheduling in edge computing. The operations of the method **1400** may be implemented by a client device (such as a UE) or its components as described herein. For example, the operations of the method **1400** may be performed by a client device or a UE **115** as described with reference to FIGS. **1-9** and **11**. In some examples, a client device may execute a set of instructions to control the functional elements of the client device to perform the described functions. Additionally, or alternatively, the client device may perform aspects of the described functions using special-purpose hardware.

[0251] At **1405**, the method may include transmitting, via a request associated with a workload, information indicative of one or more parameters associated with the workload. The operations of **1405** may be performed in accordance with examples as disclosed herein.

[0252] At **1410**, the method may include receiving information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device. The operations of **1410** may be performed in accordance with examples as disclosed herein.

Implementation Examples are Described in the Following Numbered Clauses:

[0253] Clause 1: A method for connected edge workload scheduling at a workload scheduler device, including: receiving, via a request associated with a workload, information indicative of one or more parameters associated with the workload; and transmitting an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0254] Clause 2: The method of clause 1, further including: receiving, from the first host device, information associated with the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of

one or more reliability metrics.

[0255] Clause 3: The method of any of clauses 1-2, further including: obtaining a set of scores for a set of host devices including the first host device in accordance with a respective joint evaluation associated with each host device of a set of host devices, where each score of the set of scores is for a respective host device of the set of host devices, and where the assignment of the workload to at least the first host device is in accordance with a first score for the first host device being relatively greater than one or more scores for a remainder of the set of host devices excluding the first host device.

[0256] Clause 4: The method of clause 3, further including: receiving, from each host device of the set of host devices, information associated with a respective set of one or more computation metrics associated with that host device, a respective set of one or more communication metrics associated with that host device, a respective set of one or more power supply metrics associated with that host device, and a respective set of one or more reliability metrics associated with that host device, where the respective joint evaluation associated with each host device of the set of host devices is in accordance with the information received from each host device of the set of host devices.

[0257] Clause 5: The method of any of clauses 3-4, where obtaining the set of scores for the set of host devices includes: obtaining the first score for the first host device in accordance with the first joint evaluation; and obtaining a second score for a second host device of the set of host devices in accordance with a second joint evaluation, associated with the one or more parameters, of a second set of one or more computation metrics associated with the second host device, a second set of one or more communication metrics associated with the second host device, a second set of one or more power supply metrics associated with the second host device, and a second set of one or more reliability metrics associated with the second host device, where the first score is relatively greater than the second score.

[0258] Clause 6: The method of any of clauses 3-5, where each score of the set of scores is a respective weighted sum of a respective set of one or more computation metrics, a respective set of one or more communication metrics, a respective set of one or more power supply metrics, and a respective set of one or more reliability metrics.

[0259] Clause 7: The method of clause 6, where to obtain the respective weighted sum, a first weighting is applied to the respective set of one or more computation metrics, a second weighting is applied to the respective set of one or more communication metrics, a third weighting is applied to the respective set of one or more power supply metrics, and a fourth weighting is applied to the respective set of one or more reliability metrics.

[0260] Clause 8: The method of clause 7, further including: obtaining an indication of the first weighting, the second weighting, the third weighting, and the fourth weighting in accordance with historical data associated with workload scheduling across the set of host devices or the one or more parameters associated with the workload.

[0261] Clause 9: The method of any of clauses 1-8, further including: obtaining a set of host devices from a set of multiple host devices in association with evaluating each host device of the set of multiple host devices in accordance with the one or more parameters and one or more variants of the workload, where the set of host devices includes the first host device.

[0262] Clause 10: The method of clause 9, where evaluating each host device of the set of multiple host devices includes: evaluating whether the workload is feasible for a respective host device of the set of multiple host devices in accordance with the one or more parameters, a respective set of one or more variants of the workload, and a capability of the respective host device.

[0263] Clause 11: The method of clause 10, where the capability of the respective host device is associated with one or more percentage metrics associated with an amount of resources used for one or more existing edge compute workloads at the respective host device, one or more memory usage metrics, one or more storage usage metrics, one or more power metrics at the respective host device, a software or hardware capability of the respective host device, or one or more

communication metrics associated with over-the-air signaling between the respective host device and a client device associated with the workload.

[0264] Clause 12: The method of any of clauses 9-11, further including: including the first host device in the set of host devices in accordance with the workload being feasible for the first host device, where the workload is feasible for the first host device in accordance with the one or more parameters, a first set of one or more variants of the workload, and a first capability of the first host device; and excluding a second host device of the set of multiple host devices from the set of host devices in accordance with the workload being unfeasible for the second host device, where the workload is unfeasible for the second host device in accordance with the one or more parameters, a second set of one or more variants of the workload, and a second capability of the first host device.

[0265] Clause 13: The method of any of clauses 1-12, further including: receiving, from the first host device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0266] Clause 14: The method of clause 13, further including: transmitting an indication of an updated assignment of the workload to at least a second host device in association with receiving the rejection of the assignment of the workload to the first host device.

[0267] Clause 15: The method of any of clauses 1-14, further including: receiving information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device; and transmitting, to the first host device or a first network device associated with the first host device, link configuration information that steers the first host device to the first network device from a second network device in association with the first host device processing the edge compute workloads.

[0268] Clause 16: The method of any of clauses 1-15, further including: receiving information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device; and transmitting an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with the one or more network congestion metrics.

[0269] Clause 17: The method of any of clauses 1-16, further including: receiving information indicating that a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeds a threshold preemption rate; and transmitting an indication of a suspension of workload requests in accordance with the preemption rate exceeding the threshold preemption rate.

[0270] Clause 18: The method of any of clauses 1-17, where the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0271] Clause 19: The method of clause 18, where the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources

used for the one or more existing edge compute workloads.

[0272] Clause 20: The method of any of clauses 1-19, where the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0273] Clause 21: A method for connected edge workload scheduling at a first host device, including: transmitting, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device; and receiving, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters associated with the workload, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0274] Clause 22: The method of clause 21, further including: transmitting, to a client device requesting the workload, information associated with the workload in accordance with the assignment of the workload to at least the first host device.

[0275] Clause 23: The method of clause 21, further including: transmitting, to the workload scheduler device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0276] Clause 24: The method of any of clauses 21-23, further including: transmitting, to the workload scheduler device, information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device; and receiving link configuration information that steers the first host device to a first network device from a second network device in association with the first host device processing the edge compute workloads.

[0277] Clause 25: The method of any of clauses 21-24, further including: receiving, from the workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0278] Clause 26: The method of any of clauses 21-25, further including: transmitting, to the workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0279] Clause 27: The method of any of clauses 21-26, further including: receiving, from the workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0280] Clause 28: The method of any of clauses 21-27, further including: transmitting, to the workload scheduler device, information indicating that a preemption rate at the first host device exceeds a threshold preemption rate.

[0281] Clause 29: The method of clause 28, further including: preempting one or more workloads assigned to the first host device in accordance with one or more network changes, cluster exits, higher-priority edge compute workload launches, or local application launches, where transmitting the information indicating that the preemption rate at the first host device exceeds the threshold preemption rate is in association with preempting the one or more workloads.

[0282] Clause 30: The method of any of clauses 21-29, where the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more

NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0283] Clause 31: The method of clause 30, where the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage metrics associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0284] Clause 32: The method of any of clauses 21-31, where the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0285] Clause 33: A method for connected edge workload scheduling at a client device, including: transmitting, via a request associated with a workload, information indicative of one or more parameters associated with the workload; and receiving information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0286] Clause 34: The method of clause 33, further including: receiving, from a workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0287] Clause 35: The method of any of clauses 33-34, further including: transmitting, to a workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0288] Clause 36: The method of any of clauses 33-35, further including: receiving, from a workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0289] Clause 37: The method of any of clauses 33-36, where the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0290] Clause 38: The method of clause 37, where the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or

one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage metrics associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0291] Clause 39: The method of any of clauses 33-38, where the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0292] Clause 40: A workload scheduler device, including: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the workload scheduler device to: receive, via a request associated with a workload, information indicative of one or more parameters associated with the workload; and transmit an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0293] Clause 41: The workload scheduler device of clause 40, where the processing system is further configured to cause the workload scheduler device to: receive, from the first host device, information associated with the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0294] Clause 42: The workload scheduler device of any of clauses 40-41, where the processing system is further configured to cause the workload scheduler device to: obtain a set of scores for a set of host devices including the first host device in accordance with a respective joint evaluation associated with each host device of a set of host devices, where each score of the set of scores is for a respective host device of the set of host devices, and where the assignment of the workload to at least the first host device is in accordance with a first score for the first host device being relatively greater than one or more scores for a remainder of the set of host devices excluding the first host device.

[0295] Clause 43: The workload scheduler device of clause 42, where the processing system is further configured to cause the workload scheduler device to: receive, from each host device of the set of host devices, information associated with a respective set of one or more computation metrics associated with that host device, a respective set of one or more communication metrics associated with that host device, a respective set of one or more power supply metrics associated with that host device, and a respective set of one or more reliability metrics associated with that host device, where the respective joint evaluation associated with each host device of the set of host devices is in accordance with the information received from each host device of the set of host devices.

[0296] Clause 44: The workload scheduler device of any of clauses 42-43, where, to obtain the set of scores for the set of host devices, the processing system is configured to cause the workload scheduler device to: obtain the first score for the first host device in accordance with the first joint evaluation; and obtain a second score for a second host device of the set of host devices in accordance with a second joint evaluation, associated with the one or more parameters, of a second set of one or more computation metrics associated with the second host device, a second set of one or more communication metrics associated with the second host device, a second set of one or more power supply metrics associated with the second host device, and a second set of one or more reliability metrics associated with the second host device, where the first score is relatively greater

than the second score.

[0297] Clause 45: The workload scheduler device of any of clauses 42-44, where each score of the set of scores is a respective weighted sum of a respective set of one or more computation metrics, a respective set of one or more communication metrics, a respective set of one or more power supply metrics, and a respective set of one or more reliability metrics.

[0298] Clause 46: The workload scheduler device of clause 45, where to obtain the respective weighted sum, a first weighting is applied to the respective set of one or more computation metrics, a second weighting is applied to the respective set of one or more communication metrics, a third weighting is applied to the respective set of one or more power supply metrics, and a fourth weighting is applied to the respective set of one or more reliability metrics.

[0299] Clause 47: The workload scheduler device of clause 46, where the processing system is further configured to cause the workload scheduler device to: obtain an indication of the first weighting, the second weighting, the third weighting, and the fourth weighting in accordance with historical data associated with workload scheduling across the set of host devices or the one or more parameters associated with the workload.

[0300] Clause 48: The workload scheduler device of any of clauses 40-47, where the processing system is further configured to cause the workload scheduler device to: obtain a set of host devices from a set of multiple host devices in association with evaluating each host device of the set of multiple host devices in accordance with the one or more parameters and one or more variants of the workload, where the set of host devices includes the first host device.

[0301] Clause 49: The workload scheduler device of clause 48, where, to evaluate each host device of the set of multiple host devices, the processing system is configured to cause the workload scheduler device to: evaluate whether the workload is feasible for a respective host device of the set of multiple host devices in accordance with the one or more parameters, a respective set of one or more variants of the workload, and a capability of the respective host device.

[0302] Clause 50: The workload scheduler device of clause 49, where the capability of the respective host device is associated with one or more percentage metrics associated with an amount of resources used for one or more existing edge compute workloads at the respective host device, one or more memory usage metrics, one or more storage usage metrics, one or more power metrics at the respective host device, a software or hardware capability of the respective host device, or one or more communication metrics associated with over-the-air signaling between the respective host device and a client device associated with the workload.

[0303] Clause 51: The workload scheduler device of any of clauses 48-50, where the processing system is further configured to cause the workload scheduler device to: include the first host device in the set of host devices in accordance with the workload being feasible for the first host device, where the workload is feasible for the first host device in accordance with the one or more parameters, a first set of one or more variants of the workload, and a first capability of the first host device; and exclude a second host device of the set of multiple host devices from the set of host devices in accordance with the workload being unfeasible for the second host device, where the workload is unfeasible for the second host device in accordance with the one or more parameters, a second set of one or more variants of the workload, and a second capability of the first host device.

[0304] Clause 52: The workload scheduler device of any of clauses 40-51, where the processing system is further configured to cause the workload scheduler device to: receive, from the first host device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0305] Clause 53: The workload scheduler device of clause 52, where the processing system is further configured to cause the workload scheduler device to: transmit an indication of an updated assignment of the workload to at least a second host device in association with receiving the

rejection of the assignment of the workload to the first host device.

[0306] Clause 54: The workload scheduler device of any of clauses 40-53, where the processing system is further configured to cause the workload scheduler device to: receive information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device; and transmit, to the first host device or a first network device associated with the first host device, link configuration information that steers the first host device to the first network device from a second network device in association with the first host device processing the edge compute workloads.

[0307] Clause 55: The workload scheduler device of any of clauses 40-54, where the processing system is further configured to cause the workload scheduler device to: receive information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device; and transmit an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with the one or more network congestion metrics.

[0308] Clause 56: The workload scheduler device of any of clauses 40-55, where the processing system is further configured to cause the workload scheduler device to: receive information indicating that a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeds a threshold preemption rate; and transmit an indication of a suspension of workload requests in accordance with the preemption rate exceeding the threshold preemption rate.

[0309] Clause 57: The workload scheduler device of any of clauses 40-56, where the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0310] Clause 58: The workload scheduler device of clause 57, where the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0311] Clause 59: The workload scheduler device of any of clauses 40-58, where the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0312] Clause 60: A first host device, including: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the first host device to: transmit, to a workload scheduler device, information associated with a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device; and receive, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first

joint evaluation, associated with one or more parameters associated with the workload, of the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

[0313] Clause 61: The first host device of clause 60, where the processing system is further configured to cause the first host device to: transmit, to a client device requesting the workload, information associated with the workload in accordance with the assignment of the workload to at least the first host device.

[0314] Clause 62: The first host device of clause 60, where the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

[0315] Clause 63: The first host device of any of clauses 60-62, where the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device; and receive link configuration information that steers the first host device to a first network device from a second network device in association with the first host device processing the edge compute workloads.

[0316] Clause 64: The first host device of any of clauses 60-63, where the processing system is further configured to cause the first host device to: receive, from the workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0317] Clause 65: The first host device of any of clauses 60-64, where the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0318] Clause 66: The first host device of any of clauses 60-65, where the processing system is further configured to cause the first host device to: receive, from the workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0319] Clause 67: The first host device of any of clauses 60-66, where the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, information indicating that a preemption rate at the first host device exceeds a threshold preemption rate.

[0320] Clause 68: The first host device of clause 67, where the processing system is further configured to cause the first host device to: preempt one or more workloads assigned to the first host device in accordance with one or more network changes, cluster exits, higher-priority edge compute workload launches, or local application launches, where transmitting the information indicating that the preemption rate at the first host device exceeds the threshold preemption rate is in association with preempting the one or more workloads.

[0321] Clause 69: The first host device of any of clauses 60-68, where the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0322] Clause 70: The first host device of clause 69, where the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage metrics associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0323] Clause 71: The first host device of any of clauses 60-70, where the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0324] Clause 72: A client device, including: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the client device to: transmit, via a request associated with a workload, information indicative of one or more parameters associated with the workload; and receive information associated with the workload from a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

[0325] Clause 73: The client device of clause 72, where the processing system is further configured to cause the client device to: receive, from a workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

[0326] Clause 74: The client device of any of clauses 72-73, where the processing system is further configured to cause the client device to: transmit, to a workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

[0327] Clause 75: The client device of any of clauses 72-74, where the processing system is further configured to cause the client device to: receive, from a workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

[0328] Clause 76: The client device of any of clauses 72-75, where the first set of one or more computation metrics include one or more CPU metrics, one or more GPU metrics, one or more NPU metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

[0329] Clause 77: The client device of clause 76, where the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation

resource usage metrics associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

[0330] Clause 78: The client device of any of clauses 72-77, where the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested SLA.

[0331] Clause 79: A workload scheduler device for connected edge workload scheduling, including at least one means for performing a method of any of aspects 1-20.

[0332] Aspect 80: A non-transitory computer-readable medium storing code for connected edge workload scheduling, the code including instructions executable by a processing system to perform a method of any of aspects 1-20.

[0333] Aspect 81: A first host device for connected edge workload scheduling, including at least one means for performing a method of any of aspects 21-32.

[0334] Aspect 82: A non-transitory computer-readable medium storing code for connected edge workload scheduling, the code including instructions executable by a processing system to perform a method of any of aspects 21-32.

[0335] Aspect 83: A client device for connected edge workload scheduling, including at least one means for performing a method of any of aspects 33-39.

[0336] Aspect 84: A non-transitory computer-readable medium storing code for connected edge workload scheduling, the code including instructions executable by a processing system to perform a method of any of aspects 33-39.

[0337] As used herein, the term “determine” or “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (such as via looking up in a table, a database or another data structure), inferring, ascertaining, and the like. Also, “determining” can include receiving (such as receiving information), accessing (such as accessing data stored in memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and other such similar actions.

[0338] As used herein, a phrase referring to “at least one of” or “one or more of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c. As used herein, “or” is intended to be interpreted in the inclusive sense, unless otherwise explicitly indicated. For example, “a or b” may include a only, b only, or a combination of a and b. Furthermore, as used herein, a phrase referring to “a” or “an” element refers to one or more of such elements acting individually or collectively to perform the recited function(s). Additionally, a “set” refers to one or more items, and a “subset” refers to less than a whole set, but non-empty.

[0339] As used herein, “based on” is intended to be interpreted in the inclusive sense, unless otherwise explicitly indicated. For example, “based on” may be used interchangeably with “based at least in part on,” “associated with,” “in association with,” or “in accordance with” unless otherwise explicitly indicated. Specifically, unless a phrase refers to “based on only ‘a,’” or the equivalent in context, whatever it is that is “based on ‘a,’” or “based at least in part on ‘a,’” may be based on “a” alone or based on a combination of “a” and one or more other factors, conditions, or information.

[0340] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” “at least one of one or more” may be interchangeable. For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple

components. Thus, the term “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” may refer to any or all of the one or more components. For example, a component introduced with the article “a” may be understood to mean “one or more components,” and referring to “the component” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.” Similarly, subsequent reference to a component introduced as “one or more components” using the terms “the” or “said” may refer to any or all of the one or more components. For example, referring to “the one or more components” subsequently in the claims may be understood to be equivalent to referring to “at least one of the one or more components.”

[0341] The various illustrative logics, logical blocks, modules, circuits and algorithm processes described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. The interchangeability of hardware and software has been described generally, in terms of functionality, and illustrated in the various illustrative components, blocks, modules, circuits and processes described above. Whether such functionality is implemented using hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0342] The hardware and data processing apparatus used to implement the various illustrative logics, logical blocks, modules and circuits described in connection with the aspects disclosed herein may be implemented or performed using a general purpose single- or multi-chip processor, a DSP, an ASIC, a GPU, an NPU, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, or any processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some implementations, particular processes and methods may be performed by circuitry that is specific to a given function.

[0343] In one or more aspects, the functions described may be implemented using hardware, digital electronic circuitry, computer software, firmware, including the structures disclosed in this specification and their structural equivalents thereof, or in any combination thereof.

Implementations of the subject matter described in this specification also can be implemented as one or more computer programs, such as one or more modules of computer program instructions, encoded on a computer storage media for execution by, or to control the operation of, data processing apparatus.

[0344] If implemented in software, the functions may be stored on or transmitted using one or more instructions or code of a computer-readable medium. The processes of a method or algorithm disclosed herein may be implemented in a processor-executable software module which may reside on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that can be enabled to transfer a computer program from one location to another. A storage media may be any available media that may be accessed by a computer. By way of example, and not limitation, such computer-readable media may include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer. Also, any connection can be properly termed a computer-readable medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc. Disks may reproduce data magnetically and discs may reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable

media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and instructions on a machine readable medium and computer-readable medium, which may be incorporated into a computer program product.

[0345] Various modifications to the implementations described in this disclosure may be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the claims are not intended to be limited to the implementations shown herein, but are to be accorded the widest scope consistent with this disclosure, the principles and the features disclosed herein.

[0346] Additionally, a person having ordinary skill in the art will readily appreciate, the terms “upper” and “lower” are sometimes used for ease of describing the figures, and indicate relative positions corresponding to the orientation of the figure on a properly oriented page, and may not reflect the proper orientation of any device as implemented.

[0347] Certain features that are described in this specification in the context of separate implementations also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in some combinations and even initially claimed as such, one or more features from a claimed combination can be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0348] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one more example processes in the form of a flow diagram. However, other operations that are not depicted can be incorporated in the example processes that are schematically illustrated. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the illustrated operations. In some circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products. Additionally, other implementations are within the scope of the following claims. In some implementations, the actions recited in the claims can be performed in a different order and still achieve desirable results.

Claims

1. A workload scheduler device, comprising: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the workload scheduler device to: receive, via a request associated with a workload, information indicative of one or more parameters associated with the workload; and transmit an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of: a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.
2. The workload scheduler device of claim 1, wherein the processing system is further configured to cause the workload scheduler device to: receive, from the first host device, information associated with: the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or

more reliability metrics.

3. The workload scheduler device of claim 1, wherein the processing system is further configured to cause the workload scheduler device to: obtain a set of scores for a set of host devices including the first host device in accordance with a respective joint evaluation associated with each host device of a set of host devices, wherein each score of the set of scores is for a respective host device of the set of host devices, and wherein the assignment of the workload to at least the first host device is in accordance with a first score for the first host device being relatively greater than one or more scores for a remainder of the set of host devices excluding the first host device.

4. The workload scheduler device of claim 3, wherein the processing system is further configured to cause the workload scheduler device to: receive, from each host device of the set of host devices, information associated with: a respective set of one or more computation metrics associated with that host device, a respective set of one or more communication metrics associated with that host device, a respective set of one or more power supply metrics associated with that host device, and a respective set of one or more reliability metrics associated with that host device, wherein the respective joint evaluation associated with each host device of the set of host devices is in accordance with the information received from each host device of the set of host devices.

5. The workload scheduler device of claim 3, wherein, to obtain the set of scores for the set of host devices, the processing system is configured to cause the workload scheduler device to: obtain the first score for the first host device in accordance with the first joint evaluation; and obtain a second score for a second host device of the set of host devices in accordance with a second joint evaluation, associated with the one or more parameters, of: a second set of one or more computation metrics associated with the second host device, a second set of one or more communication metrics associated with the second host device, a second set of one or more power supply metrics associated with the second host device, and a second set of one or more reliability metrics associated with the second host device, wherein the first score is relatively greater than the second score.

6. The workload scheduler device of claim 3, wherein each score of the set of scores is a respective weighted sum of: a respective set of one or more computation metrics, a respective set of one or more communication metrics, a respective set of one or more power supply metrics, and a respective set of one or more reliability metrics.

7. The workload scheduler device of claim 6, wherein to obtain the respective weighted sum: a first weighting is applied to the respective set of one or more computation metrics, a second weighting is applied to the respective set of one or more communication metrics, a third weighting is applied to the respective set of one or more power supply metrics, and a fourth weighting is applied to the respective set of one or more reliability metrics.

8. The workload scheduler device of claim 7, wherein the processing system is further configured to cause the workload scheduler device to: obtain an indication of the first weighting, the second weighting, the third weighting, and the fourth weighting in accordance with historical data associated with workload scheduling across the set of host devices or the one or more parameters associated with the workload.

9. The workload scheduler device of claim 1, wherein the processing system is further configured to cause the workload scheduler device to: obtain a set of host devices from a plurality of host devices in association with evaluating each host device of the plurality of host devices in accordance with the one or more parameters and one or more variants of the workload, wherein the set of host devices includes the first host device.

10. The workload scheduler device of claim 9, wherein, to evaluate each host device of the plurality of host devices, the processing system is configured to cause the workload scheduler device to: evaluate whether the workload is feasible for a respective host device of the plurality of host devices in accordance with the one or more parameters, a respective set of one or more variants of the workload, and a capability of the respective host device.

11. The workload scheduler device of claim 10, wherein the capability of the respective host device is associated with one or more percentage metrics associated with an amount of resources used for one or more existing edge compute workloads at the respective host device, one or more memory usage metrics, one or more storage usage metrics, one or more power metrics at the respective host device, a software or hardware capability of the respective host device, or one or more communication metrics associated with over-the-air signaling between the respective host device and a client device associated with the workload.

12. The workload scheduler device of claim 9, wherein the processing system is further configured to cause the workload scheduler device to: include the first host device in the set of host devices in accordance with the workload being feasible for the first host device, wherein the workload is feasible for the first host device in accordance with the one or more parameters, a first set of one or more variants of the workload, and a first capability of the first host device; and exclude a second host device of the plurality of host devices from the set of host devices in accordance with the workload being unfeasible for the second host device, wherein the workload is unfeasible for the second host device in accordance with the one or more parameters, a second set of one or more variants of the workload, and a second capability of the first host device.

13-17. (canceled)

18. The workload scheduler device of claim 1, wherein: the first set of one or more computation metrics include one or more central processing unit (CPU) metrics, one or more graphics processing unit (GPU) metrics, one or more neural processing unit (NPU) metrics, one or more memory metrics, or one or more storage metrics; the first set of one or more communication metrics include one or more connectivity metrics, one or more throughput metrics, one or more latency metrics, or one or more link condition metrics; the first set of one or more power supply metrics include one or more power source metrics or one or more battery power metrics; and the first set of one or more reliability metrics include one or more preemption metrics, one or more mobility metrics, or one or more usage metrics.

19. The workload scheduler device of claim 18, wherein: the one or more preemption metrics include an indication of a quantity of preemptions associated with network changes, cluster exits, higher-priority edge compute workload launches, or local application launches; the one or more mobility metrics include one or more channel variation metrics, one or more topology change metrics, or one or more cluster exit metrics; and the one or more usage metrics include one or more duration of usage metrics associated with one or more local applications, one or more computation resource usage associated with the one or more local applications, an indication of one or more existing edge compute workloads, or one or more percentage metrics associated with an amount of resources used for the one or more existing edge compute workloads.

20. The workload scheduler device of claim 1, wherein the one or more parameters associated with the workload include a workload type, an indication of an affinity of the workload, a requested completion time, a requested power source or battery power metric, a requested software or hardware capability, one or more weightings associated with the first joint evaluation, or a requested service level agreement (SLA).

21. A first host device, comprising: a processing system that includes processor circuitry and memory circuitry that stores code, the processing system configured to cause the first host device to: transmit, to a workload scheduler device, information associated with: a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device; and receive, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters, of: the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power

supply metrics, and the first set of one or more reliability metrics.

22. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: transmit, to a client device requesting the workload, information associated with the workload in accordance with the assignment of the workload to at least the first host device.

23. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

24. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload assigned to the first host device; and receive link configuration information that steers the first host device to a first network device from a second network device in association with the first host device processing the edge compute workloads.

25. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: receive, from the workload scheduler device, an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with one or more network congestion metrics.

26. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device.

27. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: receive, from the workload scheduler device, an indication of a suspension of workload requests in accordance with a preemption rate associated with workloads assigned to a cluster of edge devices including the first host device exceeding a threshold preemption rate.

28. The first host device of claim 21, wherein the processing system is further configured to cause the first host device to: transmit, to the workload scheduler device, information indicating that a preemption rate at the first host device exceeds a threshold preemption rate.

29-39. (canceled)

40. A method for connected edge workload scheduling at a workload scheduler device, comprising: receiving, via a request associated with a workload, information indicative of one or more parameters associated with the workload; and transmitting an indication of an assignment of the workload to at least a first host device in accordance with at least a first joint evaluation, associated with the one or more parameters, of: a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device.

41-51. (canceled)

52. The method of claim 40, further comprising: receiving, from the first host device, a rejection of the assignment of the workload to the first host device in accordance with one or more percentage metrics associated with an amount of resources used for one or more edge compute workloads, including the workload, at the first host device exceeding one or more threshold percentages.

53. The method of claim 52, further comprising: transmitting an indication of an updated assignment of the workload to at least a second host device in association with receiving the rejection of the assignment of the workload to the first host device.

54. The method of claim 40, further comprising: receiving information indicating that the first host device is processing edge compute workloads, the edge compute workloads including the workload

assigned to the first host device; and transmitting, to the first host device or a first network device associated with the first host device, link configuration information that steers the first host device to the first network device from a second network device in association with the first host device processing the edge compute workloads.

55. The method of claim 40, further comprising: receiving information indicative of one or more network congestion metrics associated with a cluster of edge devices including the first host device; and transmitting an indication of a suspension of workload requests associated with a session-oriented workload type in accordance with the one or more network congestion metrics.

56-59. (canceled)

60. A method for connected edge workload scheduling at a first host device, comprising: transmitting, to a workload scheduler device, information associated with: a first set of one or more computation metrics associated with the first host device, a first set of one or more communication metrics associated with the first host device, a first set of one or more power supply metrics associated with the first host device, and a first set of one or more reliability metrics associated with the first host device; and receiving, from the workload scheduler device, an indication of an assignment of a workload to at least the first host device in accordance with at least a first joint evaluation, associated with one or more parameters, of: the first set of one or more computation metrics, the first set of one or more communication metrics, the first set of one or more power supply metrics, and the first set of one or more reliability metrics.

61. The method of claim 60, further comprising: transmitting, to a client device requesting the workload, information associated with the workload in accordance with the assignment of the workload to at least the first host device.

62-78. (canceled)
